

firms name 2, the order is split and the government pays 2 per unit. If both name 1, the order is split and the government pays $x > 1$ per unit. If one names 1 and the other names 2, the order goes to the firm that named 1, and the government pays $y > 1$ per unit. What must x and y be in order to get the firms to reveal truthfully their costs? Assume that firm 1 believes that firm 2 will truthfully reveal its costs. (Assume $x, y \leq 2$ so this is valid.) Then if firm 1's costs are 1, it can name 2 and have profits of 50 with probability $1/2$. Or it can name 1 and have profits of $50(x - 1)$ with probability $1/2$ (if the other firm's costs are 1) and $100(y - 1)$ with probability $1/2$ (if the other firm's costs are 2). Assuming the firms try to maximize their expected profits, firm 1 truthfully reveals a cost of 1 if

$$\frac{1}{2}50(x - 1) + \frac{1}{2}100(y - 1) \geq 25.$$

Or, rather, it is a Nash equilibrium between the two firms to truthfully reveal their costs in this case. Let us proceed under the assumption that firms truthfully reveal their costs. Then the government's expected costs are

$$100\left[\frac{1}{4}x + \frac{1}{2}y + \frac{1}{4}2\right].$$

That is, with probability $1/4$, the government pays x per unit, with probability $1/2$, it pays y per unit, and with probability $1/4$, it pays 2 per unit. The truth-inducing constraint can be written $25x + 50y \geq 100$, and the government's objective function is to minimize $25x + 50y + 50$, so it is evident that any selection of x and y that satisfies the constraint with equality gives minimum expected cost to the government, and that expected cost is 150.

Suppose the government chooses $x = 2$ and $y = 1$. Is truth-telling the only Nash equilibrium? Suppose firm 2 reports costs of 2 regardless of what its costs are. If firm 1 reports costs of 2 when its costs are in fact 1, then it is sure to make 50. (It gets half the order for sure, and it is paid 2 per unit when its costs are 1.) If it reports costs of 1 when its costs are 1, it gets the entire order but makes nothing. (It has reported 1, its rival has reported 2, and so it is paid precisely its costs.) Of course, when its costs are 2 it will name 2 as its costs. Hence if firm 2 always reports costs of 2, then it is a best response for firm 1 always to report costs of 2. It is a Nash equilibrium for both firms always to report costs of 2, and at this Nash equilibrium, each firm has expected profits of 25. By comparison, at the Nash equilibrium where both firms always tell the truth, each has

expected profits of 12.5. If we believe that firms will find their way to Nash equilibria that are better for both of them, then $x = 2$ and $y = 1$ doesn't seem so good. At least, the government is risking the possibility that the firms find a Nash equilibrium that is better for both of them and worse for the government.

To avoid this, we clearly want to make y as large as possible. So consider setting $x = 0$ and $y = 2$. (It is assumed that x and y are constrained to be nonnegative. Note that $x = 0$ is bad enough; the firms are told by the government that if each reports low costs, each gets to provide half the order for free.) If $x = 0$ and $y = 2$, and if firm 2 always reports costs of 2, then firm 1, when its costs are 1, nets 50 for sure by reporting costs of 2, and 100 for sure by reporting costs of 1. Hence we would not have the problem of an equilibrium where both firms report 2 regardless of their costs. But now something worse comes along. Suppose one firm adopts the strategy of always reporting that its costs are 1. The other firm, in response, can never do better than to report costs of 2; if it reports costs of 1, it will have to produce half the output and it will be paid nothing. And if the second firm always reports costs of 2, then among the first firm's best responses is always to report costs of 1; this gives it 2 per unit regardless of its costs. We have a second Nash equilibrium with an expected cost to the government of 200.

So y is too high. Try $y = 1.9$ and $x = .2$. Now we're in business. If a firm's costs are 2, it will not report 1, no matter what it thinks the other firm is doing. If it reports 1, it will be saddled with some of the output, and it won't recover its costs. While if a firm's costs are 1, it must assume that its rival will report 2 when its costs are 2. If it thinks its rival will truthfully report 1 when its costs are 1, then it is indifferent between reporting 1 or 2. If it thinks its rival will report 2 when its costs are 1, then it strictly prefers to report costs of 1; reporting 2 nets 50, while reporting 1 nets 90. If we sweeten the pot just a bit, making $y = 1.91$ and $x = .21$, then the unique Nash equilibrium between the two firms, given the mechanism the government has imposed, is to reveal costs truthfully. And the government's expected cost of procurement is 150.50. (The extra .50 is the result of sweetening the pot.)

It is possible, though, to continue to quibble about this scheme. Suppose that firm 1 gets it into its head that firm 2 will always report a cost of 1, even though this is not Nash behavior. If firm 1's costs are 1, then reporting the truth means that firm 1 will sustain a loss (it gets half the order and is paid less than its costs). So firm 1, if it is convinced that firm 2 will always report a cost of 1, prefers to name costs of 2 (or to refuse to participate altogether).

Consider the following scheme. The government announces that if both firms claim that their costs are 2, the order will be split between them, and each will be paid 2 per unit. If both claim that their costs are 1, the order will be split between them and each will be paid 1.01. And if one claims that its costs are 1 and the other claims its costs are 2, then the one claiming costs of 1 will get the entire order and will be paid 1.51 per unit. And each firm will be given .01 for participating at all.

Suppose firm 1 has costs of 2. Then no matter what it thinks firm 2 will do, it is better to name 2 than to name 1, and the .01 participation bonus means that it is better to name 2 than to refuse to participate. Suppose firm 1 has costs of 1. If firm 2 names costs of 1, it is better for firm 1 to name a cost of 1 (and make .01 on each of fifty units) than to name a cost of 2 and make nothing. If firm 2 names costs of 2, it is better for firm 1 to name a cost of 1 than a cost of 2; naming 2 will net 1.00 on each of fifty units, while naming 1 will net .51 on each of one hundred units. That is, against this scheme, truth-telling is a strictly dominant strategy for each of the firms. Note that this costs the government $(1/4)101 + (1/2)151 + (1/4)200 + .02 = 150.77$. Moreover, by making the sweeteners (the extra .01's) vanishingly small, the cost falls to 150.

Let us summarize where we are. We found a scheme in which truth-telling is a Nash equilibrium, costing the government an expected 150. But this scheme has multiple Nash equilibria for the two firms, one of which is better for both firms than is truth-telling. So we modified the scheme in a way that makes truth-telling the unique Nash equilibrium. But truth-telling isn't in this case, a dominant strategy. So we modified the scheme again, so that truth-telling becomes a dominant strategy for each firm. And still, except for "sweeteners" that can be made arbitrarily small, the expected cost to the government is 150.

Optimal direct revelation mechanisms

In the foregoing analysis, we made two special assumptions about the mechanism the government employs to award the contract. First, we made a qualitative assumption that the government would ask the firms to name their costs, with the contract amounts and corresponding payments depending on the pair of named costs. Second, we assumed that the contract amounts and corresponding payments took a particular form. The contract quantities are split if the firms name the same price, and the entire contract is given to the firm naming the lower price. If both name 2, each is paid 2 per unit. If one names 1 and the other 2, the one naming 1 is paid y per unit. And if both name 1, both are paid x per unit.

If we look at different schemes, either of the same general qualitative structure but with different sorts of quantity assignments and payments, or of a completely different form, perhaps we can find one that will lower the government's expected costs below 150. Over the next two subsections we will see that this can't be done, subject to some qualifications. In this section, we keep to the general qualitative form used previously, varying the parameterization. In the next section, we argue that *any* general mechanism is "equivalent" in terms of outcomes to some mechanism of the qualitative form of this section. The second step, which is called the *revelation principle*, gives this chapter its name.

We begin with a definition of the "general qualitative structure" of the mechanisms discussed in the previous subsection. A *direct revelation mechanism* is any mechanism of the following sort. For each of the four pairs mn such that $m = 1, 2$ and $n = 1, 2$, there is given a four-tuple $(x_{mn}^1, x_{mn}^2, t_{mn}^1, t_{mn}^2) \in R_+^4$ such that $x_{mn}^1 + x_{mn}^2 \geq 100$. (By R_+^4 , we mean the nonnegative orthant in R^4 .) The two firms, who already know their true costs, are given the opportunity to participate. If they elect to do so, they simultaneously and independently announce whether their costs are 1 or 2. (This is what gives a direct revelation mechanism its name.) Letting m be the announcement of firm 1 and n the announcement of firm 2, the four-tuple $(x_{mn}^1, x_{mn}^2, t_{mn}^1, t_{mn}^2)$ corresponding to their announcement mn determines their responsibilities: Firm i must produce x_{mn}^i units of the item, for which it receives a payment of t_{mn}^i . Note that if c^i is the true marginal cost of firm i , then the firm's net profits are $t_{mn}^i - c^i x_{mn}^i$.

Note carefully that we have allowed firms the option of refusing to participate at the stage after they know their true costs but before they have learned what the other firm announces. In effect, we are supposing that the government can compel the firms to produce according to the "rules" of the mechanism if they agree to participate at the point indicated. We can think of at least two other possibilities: (a) The government cannot compel the firms to carry out the dictates of the mechanism at any time; firms need only carry through with the dictates of the mechanism if they continue not to expect a loss; (b) the government can propose this mechanism to the two firms before they know their costs, and they are compelled to carry through with the mechanism if they agree at that point.

In restricting the t^i to be nonnegative, we are making a further assumption, namely that the government cannot compel the firm to make a payment to the government. You may feel that this is in the spirit of but a poor substitute for (a); if the idea is that the government cannot compel the firm to take a loss ex post, then (a) and not our nonnegativity constraint would seem appropriate. We continue our analysis under the assumptions that the firm can be compelled to take a loss if it agrees to participate after it learns its own cost but cannot be compelled to make a monetary payment to the govern-

ment, bearing in mind that perhaps the sign of the t^i should be unrestricted or that (a) or (b) should be assumed.

Now we ask the following question.

Suppose the government restricts attention to direct revelation mechanisms with the property that agreeing to participate and then truthfully revealing costs constitutes a Nash equilibrium for the two firms in the corresponding game of incomplete information between them. What is the lowest expected payment that government must make to the firms using such a mechanism?

You may wonder why we are asking (and answering) this question. If so, suspend your wonder for the time being.

This question can be turned into a linear programming problem. A linear programming problem is a constrained maximization problem with linear objective function and linear constraints. Such problems are very easily solved numerically, using a variety of methods, the most prominent of which are variations on the *simplex algorithm*. If you don't know about linear programming, you should learn; Luenberger (1984) gives an accessible treatment. In any event, in all the optimization problems that follow in this chapter, we will announce answers. This may seem a bit like magic if you don't know about linear programming, but in fact we are simply reporting solutions obtained by the simplex algorithm. (Trying to solve the problems given below by hand with Lagrange multipliers is *not* recommended.)

To begin, we consider what is required if participation and truth-telling are to be a Nash equilibrium for a given direct revelation mechanism. For each of the two firms, we have four constraints. For firm 1 they are:

(P1-1) If firm 2 participates and tells the truth, and if firm 1's costs are 1, then firm 1 is willing to participate in the mechanism and tell the truth instead of refusing to participate,

$$\frac{1}{2}[t_{11}^1 - 1x_{11}^1] + \frac{1}{2}[t_{12}^1 - 1x_{12}^1] \geq 0.$$

(P1-2) If firm 2 participates and tells the truth, and if firm 1's costs are 2, then firm 1 is willing to participate in the mechanism and tell the truth rather than refusing to participate,

$$\frac{1}{2}[t_{21}^1 - 2x_{21}^1] + \frac{1}{2}[t_{22}^1 - 2x_{22}^1] \geq 0.$$

(I1-1) If firm 2 participates and tells the truth, and if firm 1's costs are 1, then firm 1 is willing to participate in the mechanism and tell the truth instead of participating and falsely claiming that its costs are 2,

$$\frac{1}{2}[t_{11}^1 - 1x_{11}^1] + \frac{1}{2}[t_{12}^1 - 1x_{12}^1] \geq \frac{1}{2}[t_{21}^1 - 1x_{21}^1] + \frac{1}{2}[t_{22}^1 - 1x_{22}^1].$$

(I1-2) If firm 2 participates and tells the truth, and if firm 1's costs are 2, then firm 1 is willing to participate in the mechanism and tell the truth instead of participating and falsely claiming that its costs are 1,

$$\frac{1}{2}[t_{21}^1 - 2x_{21}^1] + \frac{1}{2}[t_{22}^1 - 2x_{22}^1] \geq \frac{1}{2}[t_{11}^1 - 2x_{11}^1] + \frac{1}{2}[t_{12}^1 - 2x_{12}^1].$$

If you go through each of these four constraints carefully, you will be able to make sense of them. Take (I1-2), a mnemonic for the Incentive constraint for firm 1 when its costs are 2. On the left-hand side we have the expected profits of this firm if it participates, truthfully says that its costs are 2, and the other firm participates and tells the truth. This is the appropriately weighted average of its profits if the other firm's costs are 1 (and are so revealed) and if the other firm's costs are 2. On the right-hand side we have the expected profits of this firm when its costs are 2 if it participates and falsely maintains that its costs are only 1. Note that on both sides, the production quantity the firm is assigned is premultiplied by 2, since these are the true costs of the firm in this case.

There are four similar constraints for firm 2. We also have nonnegativity constraints (by assumption) on all the variables and four constraints that say that the government obtains its 100 units, constraints such as

$$x_{11}^1 + x_{11}^2 \geq 100.$$

The set of direct revelation mechanisms, specified by the sixteen variables x_{mn}^i and t_{mn}^i for $i, m, n = 1, 2$, that satisfy all these constraints is precisely the set of direct revelation mechanisms for which participation and truth-telling is a Nash equilibrium. So to answer the question that has been posed, we wish to minimize

$$\frac{1}{4}[t_{11}^1 + t_{11}^2] + \frac{1}{4}[t_{21}^1 + t_{21}^2] + \frac{1}{4}[t_{12}^1 + t_{12}^2] + \frac{1}{4}[t_{22}^1 + t_{22}^2],$$

which is the expected amount the government pays out in a given mechanism, subject to all the constraints so far given.

We asked the computer to solve this linear programming problem (minimize the objective function subject to all the constraints above), and it gave back: $x_{11}^1 = x_{12}^1 = x_{21}^2 = x_{22}^2 = 100$, $t_{12}^1 = t_{21}^2 = t_{22}^2 = 200$, and all other variables equal to zero. That is, firm 1 is assigned all the production if it announces costs of 1; otherwise firm 2 is assigned to produce all 100 units. Firm 1 is paid 200 if it announces its costs are 1 and firm 2 announces costs of 2, and firm 2 is paid 200 if firm 1 announces that its costs are 2. This costs the government an expected 150.

As we know from our earlier discussion, this is not the only way to induce truth-telling as a Nash equilibrium in a direct revelation game. Another way is to set $x_{11}^1 = x_{11}^2 = x_{22}^2 = 50$, $x_{12}^1 = x_{21}^2 = 100$, $t_{11}^1 = t_{11}^2 = 50$, $t_{22}^2 = t_{22}^2 = 100$, $t_{12}^1 = t_{21}^2 = 150$, and all other variables equal to zero. (This corresponds to paying 1 per unit when both firms name costs of 1, 2 per unit when both name costs of 2, and 1.5 per unit to the firm that names 1 if one names 1 and the other 2.) We will wish to refer to this particular scheme later on, so we give it a name; this will hereafter be referred to as the *nice* direct revelation scheme. The solution given in the previous paragraph is much more “extreme”; it has many more zero elements. In general, linear programming will tend to produce extreme solutions of this sort. But the point is not so much the solution produced as it is the value of the solution: There are many direct revelation schemes for which participation and truth-telling give a Nash equilibrium in this situation and which cost the government an expected 150. But there is *no* scheme for which participation and truth-telling is a Nash equilibrium that produces a lower expected cost; this is what we learned by solving the constrained optimization problem using linear programming.

In the formulation above, we have limited attention to direct revelation mechanisms where, for each pair of reports mn by the two firms, a single four-tuple $(x_{mn}^1, x_{mn}^2, t_{mn}^1, t_{mn}^2)$ is implemented. For later purposes, we need to generalize this a bit and consider direct revelation mechanisms where, for each pair mn , there is a probability distribution μ_{mn} over such four-tuples. The idea is that firms report their costs, and then a lottery over four-tuples is conducted, with distribution given by μ_{mn} for the reported pair mn . The outcome of this lottery is then implemented.

In the context of this problem, allowing for random direct revelation mechanisms doesn't change anything. Since the government and firms are risk neutral and the firms have linear production technologies, replacing any lottery μ_{mn} by the four-tuple that is its mathematical expectation doesn't affect any party's expected profits. Hence the best the government can do (in terms of minimum expected costs) with a random direct revelation mechanism (which induces participation and truth-telling as a Nash

equilibrium) is the best it can do with a nonrandom mechanism, and that is expected costs of 150.

We will return to this basic formulation in a bit, but before doing so we consider three alternative formulations, based on different notions of which constraints bind on the government in designing a direct revelation mechanism.

(a) Imagine that the government cannot compel a firm to produce if the firm would, by producing, sustain a loss. In other words, firms are allowed to refuse to participate after they learn their assignments and monetary transfers. We assume that firms, if indifferent between participating or not, will agree to participate. (This is the usual problem of weak versus strict inequalities in this sort of analysis.)

Suppose the government restricts attention to direct revelation mechanisms with the property that truthfully revealing costs and then, ex post, agreeing to participate constitutes a Nash equilibrium in the game with incomplete information between the two firms. What is the lowest expected payment that government must make to the firms in such mechanisms?

We already know the answer to this question; the lowest expected payment will be 150. We know that this is the answer by the following chain of logic: (1) The lowest expected payment under these constraints will be no lower than the lowest expected payment if a firm can be compelled to produce if it agrees to participate once it knows its own costs but not those of its rival. This is so because in any scheme where the firms will always agree to participate ex post, they will surely agree at the earlier stage. Mechanisms that meet the test of this question meet the test of the earlier question, and the minimum cost under the earlier constraints was found to be 150. Hence 150 is a lower bound on what can be achieved under these tighter restrictions.

(2) In the *nice*^b direct revelation scheme, which has expected costs to the government of 150, these extra constraints are met.

Even though we know the answer to this question, it is instructive to see how we would modify our constrained optimization problem to encompass this changed formulation. It is easy. In place of the four participation constraints we had before, we now have eight. A typical one of these is

$$t_{21}^1 - 2x_{21}^1 \geq 0, \quad (P1-21)$$

which expresses the constraint that if firm 1's costs are 2 and firm 2's costs are 1, then firm 1 is willing to undertake the assigned production level x_{21}^1 , based on a comparison of what the government will pay and what this level of production will cost.^c If we drop the four old participation constraints in

^b Go back six paragraphs if you don't know what *nice* means in this context.

^c If you write down the corresponding constraint for firm 1 when its costs are 2 and those of its rival are 2, and you compare these two constraints with the constraint (P1-2) from before, you will see why this formulation is theoretically more restrictive than the earlier formulation.

favor of these eight "ex post" participation constraints, we can again feed the problem to a linear programming algorithm. When we did this, the linear programming algorithm used didn't come up with the nice direct revelation mechanism but instead suggested the following mechanism. Everything depends on what firm 1 reports. If firm 1 reports that its costs are 1, then we pay it 100 and tell it to produce one hundred units. (We pay nothing to firm 2.) But if firm 1 reports costs of 2, we pay it nothing, we pay 200 to firm 2, and we assign firm 2 production of all one hundred units.

(b) Going in the other direction, suppose the government can approach the firms with a mechanism before they learn their costs. Firms agree whether to participate based on their initial assessments of what their costs will be, and once they agree to participate, they are compelled to complete any assignments made in exchange for compensation agreed to ex ante as part of the mechanism originally proposed. In this case, we would replace the four original participation constraints with two "ex ante" participation constraints. The constraint for firm 1 would read

$$\frac{1}{4}[t_{11}^1 - 1x_{11}^1] + \frac{1}{4}[t_{12}^1 - 1x_{12}^1] + \frac{1}{4}[t_{21}^1 - 2x_{21}^1] + \frac{1}{4}[t_{22}^1 - 2x_{22}^1] \geq 0.$$

We can ask

What is the lowest expected cost to the government if it can employ a direct revelation mechanism where truth-telling is a Nash equilibrium and where participation constraints are as just above?

If we look for a solution to this formulation, we find that costs to the government can be driven down to the first-best level of 125. One scheme that does this is: Firm 1 produces all one hundred units if it names costs of 1 or if firm 2 names costs of 2. Only if firm 1 says its costs are 2 and firm 2 says its costs are 1 does firm 2 get to produce. Firm 2 is paid 100 if it says its costs are 1 and firm 1 says its costs are 2; note that this induces truth-telling on the part of firm 2, since to claim costs of 2 means firm 2 is shut out entirely, but to claim costs of 1 when its costs are in fact 2 means it may lose money (if firm 1 claims costs of 2). Firm 1 is paid 250 if it claims costs of 1 and firm 2 claims costs of 2, and it is paid 150 if it claims costs of 2 and firm 2 claims costs of 2. Otherwise firm 1 is paid nothing. Why does this induce truth-telling by firm 1? If its costs are 1, by claiming costs of 1 it knows it will produce one hundred units and it will get 250 with probability 1/2. If it claims costs of 2, it will produce one hundred units with probability 1/2 and receive 150 with probability 1/2. Telling the truth is just as good as lying. But if its costs are 2, telling the truth means a loss of 25 (with probability 1/2 it will be told to produce one hundred, costing 200, and it gets back 150), while lying means a loss of 75 (it will be told to produce one hundred, costing 200, for sure, and it gets 250 with probability 1/2).

Variations (a) and (b) turn on when the firm is allowed to decide whether to participate or, more precisely, when the firm is no longer allowed to opt

out of a mechanism earlier agreed to. Can the firms be approached before they learn their true costs and then be compelled to continue if they agreed ex ante? Can they only be approached after they learn their true costs, or compelled to continue only after they learn those costs? Or do firms always retain the right to opt out? In the literature you will find reference to *ex ante*, *interim*, and *ex post* implementation, referring (roughly) to variations (b), the original formulation, and variation (a), respectively.

(c) As noted earlier, we have constrained the government to make nonnegative payments to the firm. We might wonder what can be achieved (say, in our original formulation) if we allow the $t_{m,n}^i$ to be negative. This change, in this case, doesn't amount to much. The minimum cost remains 150, and in fact the computer returned the same solution as we obtained for the original formulation. (However this variation will become interesting in problem 8.)

The revelation principle

We now know that the government cannot do any better than an expected cost of 150 by using a direct revelation mechanism that induces participation and truth-telling as a Nash equilibrium. Can it do better with some more complex sort of mechanism?

The *revelation principle* says that it cannot, subject to some important caveats. Truth-telling in a direct revelation mechanism (which induces truth-telling) mimics every possible *equilibrium outcome* in every possible mechanism, however complex.

The revelation principle is one of those things that is obvious once you understand it but somewhat cumbersome to explain. We try here to strike a balance between excessive formality and verbal arm waving, but the appropriate level of formality is highly subjective, and if, after reading about this here, you don't find the basic idea rather transparent, then you should seek a different treatment.

To begin, we have to say what we mean by a general mechanism. We have in mind some "game form" that the government will design and in which it asks the two firms to participate. We think of this general mechanism as a finite extensive form tree for the two firms, possibly incorporating moves by nature.^{d e} At the end of each branch of the tree is a four-tuple of the sort in the previous section. The government designs and presents this game tree to the two firms and asks if they would like to play. (We assume that each firm knows its costs but not those of its rival when faced with the decision whether to participate; variations in the spirit

^d The government is presumed able to conduct any needed randomizations objectively and credibly.

^e We assume that the game tree prescribed is one of perfect recall and the two firms are capable of playing the game tree as it is designed.

of those in the previous section are straightforward.) If both agree, then they play the game, and the four-tuple at the end of the branch they play determines what each is asked to produce and how much each is paid.

Because the government is allowed to present any finite game tree at all (with the sorts of endpoints we've described) for its mechanism, this formalism encompasses many mechanisms one can imagine.^f

In the context we have set, for every general mechanism of this sort there is a corresponding game of incomplete information for the two firms. Each firm begins with private information about its costs; think of four possible initial nodes. Append to each of these initial nodes a copy of the mechanism, and at each endpoint use the four-tuple specified by the mechanism with the costs of the firms as specified by the initial node to compute payoffs for the players. Right at the start of the game, give each player the option to decline to participate, with payoff zero for each if either declines.^g Information sets are used to mimic the idea that the firms know their own costs but not the costs of the other firm while playing this game.

An example may help clarify. Imagine the following simple mechanism. The government goes to firm 1 and offers to firm 1 the opportunity to produce the hundred units for a payment of 100. If firm 1 declines, the government goes to firm 2 and offers firm 2 the opportunity to produce the hundred units for a payment of 150. If firm 2 declines, the government goes back to firm 1 and offers it the opportunity to produce the hundred units for a payment of 200. And if firm 1 refuses, the rules specify that firms 1 and 2 each produce 50 at a payment of 10 for each firm.

We depict this mechanism in figure 18.7(a). At the end of each endpoint is a four-tuple, which gives in order the production quantity assigned to firm 1 and then to firm 2, and then the payments made to the firms.

^f Certain relatively free-form bargaining schemes may not be reducible to a definite game tree, and you may wish to consider how what we will say here applies to such situations. In this regard, note especially our later comments about games so complex that it is inappropriate to assume that players will play a Nash equilibrium.

^g If one firm accepts and the other declines, why is the payoff zero for the accepting firm? Why doesn't the government continue to conduct business with the firm that accepts? We could specify something more interesting for cases where one firm accepts and the other declines, but whatever is specified will be irrelevant to what follows, because we assume that both firms anticipate that the other will accept. You might worry that this will preclude mechanisms that are intended to ignore one firm entirely. But we can construct mechanisms where one firm or the other gets nothing and does nothing at the end of each branch; such a mechanism would be as if the government decided to deal with the other firm only, and the first firm in this mechanism would be indifferent between accepting and rejecting. Hence the revelation principle encompasses mechanisms where the government sets out to deal with one firm only.

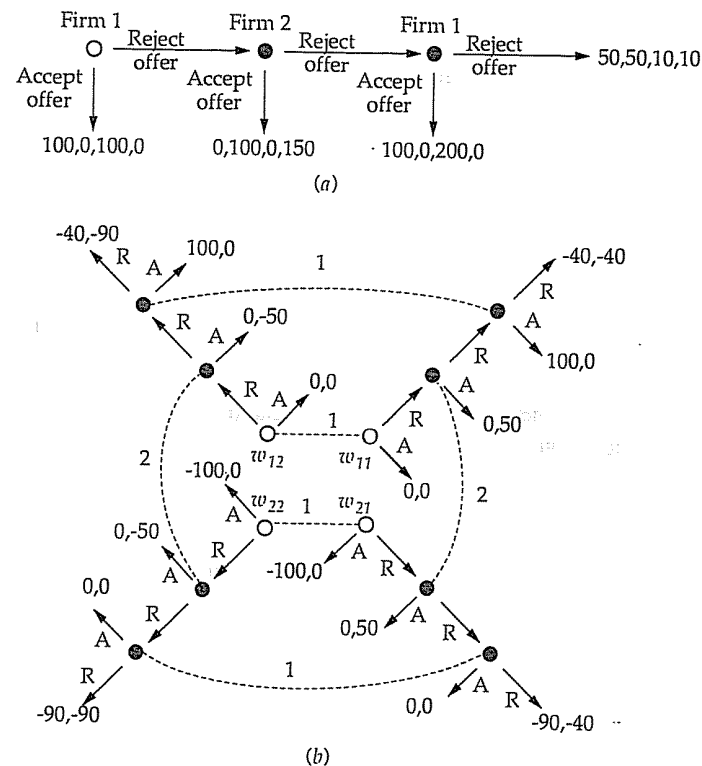


Figure 18.7. A complex mechanism and the corresponding game of incomplete information.

In (a), a complex mechanism is depicted. The endpoints give firm 1's production quantity, then 2's quantity, then 1's income, and then 2's income. In (b), we show the corresponding game of incomplete information between the players, although we have left off the diagram the opportunity to opt out at the start of the game. The four initial nodes in the game tree are labeled w_{mn} where m is the actual costs of firm 1 and n is the actual costs of firm 2. We have suppressed the initial probability distribution, which is $1/4$ on each of the four initial nodes. Endpoints are evaluated in terms of firm 1's profits and then firm 2's profits.

Then in figure 18.7(b), we construct the associated game of incomplete information.⁶ Here the endpoints are evaluated using the firms' costs as determined by the initial node in the game tree.

***The revelation principle.** Consider any mechanism of the sort described above, the associated game of incomplete information between the two firms, and any Nash equilibrium of this game in which both firms agree to participate. The expected costs to the government and the assignment of production levels and payments to the firms in this equilibrium, viewed as a function of their costs, can be precisely obtained by participation and truth-telling in a direct revelation game for which participation and truth-telling give a Nash equilibrium.*

The proof runs along the following lines. Take the mechanism, associated game, and Nash equilibrium. Think of the equilibrium strategies for the two firms as being given in the following form: Firm 1 follows (possibly mixed) strategy σ_1^1 in the mechanism tree if its costs are 1, while if its costs are 2, it follows strategy σ_2^1 ; similar notation is used for firm 2.

Now construct a (possibly random) direct revelation game as follows. Take each of the four possible pairs of costs in turn. For the pair corresponding to the costs m for firm 1 and n for firm 2, construct the probability distribution over outcomes in the mechanism tree that results from firm 1 playing σ_m^1 and firm 2 playing σ_n^2 . This gives a probability distribution over four-tuples, which in the direct revelation game is μ_{mn} .

The claim is that in this direct revelation game, truth-telling and participation is a Nash equilibrium that induces the same outcomes as the Nash equilibrium in the original game. The reason is simple (but hard to say). Suppose, by way of contradiction, that firm 1 strictly prefers in the direct revelation game to report costs of 2 when its costs are 1. By construction if firm 1's costs are 1, then the distribution over outcomes of claiming costs of 1 in the direct revelation game against truth-telling by firm 2 is precisely the same as the distribution over outcomes of playing σ_1^1 against firm 2's strategy in the original game, and the distribution over outcomes of claiming costs of 2 in the direct revelation game against truth-telling by firm 2 is precisely the same as the distribution of outcomes from playing σ_2^1 against firm 2's strategy in the original game. If firm 1, when its costs are 1, prefers to claim costs of 2 in the direct revelation game, then it must prefer to play σ_2^1 when its costs are 1 in the original game. But since we suppose that we began with a Nash equilibrium in the original game, this is impossible.

⁶ To keep the figure from becoming too complex, we have left off the initial option not to participate.

If that didn't penetrate, try putting it this way: In the original game, a firm can always act as if its costs were "the other" value. The statement that we have a Nash equilibrium means that it has no positive incentive to do this. Our direct revelation game is constructed essentially as follows. The two firms simultaneously and independently report their costs to some disinterested third party. This third party then implements in the original general mechanism what the firms would have done in the original equilibrium as a function of the costs they report. That is, if firm 1 reports costs m and firm 2 reports n , the third party implements play of σ_m^1 against play of σ_n^2 . Since a firm would not choose in the original game to act as if its costs were other than what they are, so the firm would not wish to have the third party "misplay" on its behalf.

Similar arguments ensure that firms will participate in the direct revelation game. If a firm prefers in the direct revelation game not to participate (against truth-telling by its opponent), then it would prefer not to participate in the original game (against its opponent's strategy). But the hypothesis of the revelation principle is that the firms choose to participate in the original game.

We employed the revelation principle, albeit in somewhat degenerate form, in section 18.1, when we asserted that to find the optimal marketing strategy for the insurance firm, it suffices to look for the optimal menu of contracts to offer, with the choice of contract from the menu left to the consumer. Think of implementing this choice as follows: The insurance company says that contract A will be given to any consumer who reveals that she is high risk, and B will be given to any consumer who reveals she is low risk; then the consumer is told to state which she is. Anything that can be achieved by a complex marketing scheme, the revelation principle asserts, can be asserted by this simple direct revelation mechanism. This isn't precisely the same as having the consumer choose directly between A and B, which is what we imagine when we think of a menu of choices from which the consumer chooses. But in a one-person problem, it comes down to the same thing.

A sometimes heard paraphrase of the revelation principle (adapted to this context) is: Anything that the government can achieve with a general mechanism can be achieved by a direct revelation mechanism where truth-telling is a Nash equilibrium. This, like many paraphrases, is too broad. The antecedent in the paraphrase should be: Anything that the government can achieve in a Nash equilibrium between the two firms in a general mechanism.... Getting the antecedent right is important on at least two grounds, both having to do with the relevance of Nash analysis in general mechanisms.

In the first place, the government might be able to devise a mechanism so complex that the firms do not see their way to a Nash equilibrium in it, and their behavior may then give the government lower expected costs than it gets in a truth-telling equilibrium of a direct revelation game. (In the one-person case of section 18.1, this is just the problem of the insurance-buying consumer who is not completely aware of all the choices she will be offered. Rational maximization in a one-person choice problem is “Nash equilibrium analysis” if we think of the choice problem as a one-person game.) Of course, unless we have a theory about how the firms will act in the face of mechanisms so complex that they don’t see their way to a Nash equilibrium, it will be hard to see how the government can evaluate such mechanisms. This isn’t to say that interesting theories of this sort can’t be devised. But such theories are not very well developed at present.

The second objection relates to chapter 14. We can imagine contexts in which the two firms interact repeatedly. Think of the firms as being, say, Boeing and Airbus, and the hundred units being commercial aircraft. Although perhaps not descriptive of the current relations between Boeing and Airbus, we can imagine that the firms are engaged in implicit collusion, sustained by the usual sort of folk theorem construction. If this is so, it is not entirely reasonable to suppose that the firms in one particular encounter will behave according to the dictates of Nash equilibrium in that encounter. We might anticipate two implicitly collusive firms maintaining that each has costs of 2, even if one or the other has costs of 1, and even if it is to the short-run advantage of the firm with costs of 1 to so state, because the short-run advantage must be weighed against long-run considerations of future encounters between the firms. In such a situation, the party designing the mechanism might do well to think about how to design a mechanism that *reduces* the firm’s abilities to collude, for example, by adding noise to the outcome. Direct revelation mechanisms may be more or less conducive to collusion between the firms; insofar as direct revelation supports collusion, the government may do better with a more complex mechanism.

One should be clear about the uses to which the revelation principle is put. It can be thought of as a statement about *how* actually to implement contracts. But it may be better to use it with greater circumspection as a tool of analysis for finding the limits of *what* outcomes can be implemented, without reference to how best to implement a particular outcome.⁷

We can make this point in another way. In some contexts of direct revelation, there will be situations *ex post* where the party in the role of

⁷ And, in the literature, this is primarily how it is used.

the government knows that it can obtain further gains from trade from one or more of the parties who participated. We already saw an example in the context of insurance with adverse selection, where the insurance company knows *ex post* that it can improve its profits and the welfare of the consumer, if the consumer self-identifies as being low-risk. As we said before, the insurance company must be able to restrain itself from renegotiating with a consumer who identifies herself as being low-risk. Similarly, in many applications of the revelation principle, the party in the role of mechanism designer must be able to commit credibly to no subsequent (re)negotiation once it learns the types of the parties with which it is dealing. As a matter of practical implementation, the ability to make such a credible commitment may be greater if the optimal mechanism is implemented by some scheme other than direct revelation, because more complex mechanisms may give less information (on which to base renegotiation) to the party designing the mechanism. So, to repeat, the revelation principle can be useful for saying what is the range of (conceivably) implementable outcomes, without being taken quite seriously as a statement of how to implement a given outcome.

What do we know, then, about our problem of the government procuring the one hundred units? The best that it can do (in terms of minimizing its costs) with truth-telling in a direct revelation mechanism is to get costs down to 150. If it can do better with some fancy mechanism, then this will have to be a mechanism that so confuses the two firms that they don’t play to a Nash equilibrium. In any mechanism where they play to a Nash equilibrium, the expected payments to them can be no less than 150.

Multiple equilibria and dominant strategy implementation

To continue on the theme of *how* to implement a particular outcome, we note once more that in many direct revelation mechanisms truth-telling is a Nash equilibrium that costs the government 150 in expectation, and some are “better” than others. Recall that we described a direct revelation mechanism in which truth-telling was a Nash equilibrium (with expected costs of 150), but another Nash equilibrium (involving some lying) was better for both firms and worse for the government. And we described a direct revelation mechanism in which truth-telling was the only Nash equilibrium (with expected costs of 150), but where a firm would deviate from truth-telling if it thought its rival was playing a non-Nash strategy. Best of all the mechanisms we constructed was the *nice*⁸ mechanism, in which truth-telling was a dominant strategy for each of the firms.

⁸ See footnote b if you’ve forgotten what this means.

More generally, when one designs a mechanism, either a direct revelation mechanism in which truth-telling is a Nash equilibrium or something more general, it is natural to worry whether there are other Nash equilibria (with outcomes that are not so good for the mechanism designer), or whether the participants in the mechanism will find their way to the Nash equilibrium. Everything else held equal, it seems natural to prefer a mechanism in which the equilibrium the designer wishes is the only Nash equilibrium or, at least, in which all the equilibria are no worse for the designer than the one being aimed for. Moreover, the mechanism designer might worry that the players won't settle on an equilibrium at all, for example, because the environment is too complex for players to find their way to an equilibrium.

These worries are ameliorated if the desired behavior constitutes a dominant strategy for all the players in the mechanism that is designed. If the desired behavior is a strictly dominant strategy for each player, then it is of course the unique Nash equilibrium. And more: In cases where it may be unreasonable to expect players to find their way to a Nash equilibrium, it may be reasonable to expect them to recognize (and play) a dominant strategy. Even if the strategies are not strictly dominant (so there may be other equilibria in weakly dominated strategies), the mechanism designer may feel relatively secure in a prediction that players will settle on strategies that are dominant.

These considerations motivate the notion of a *dominant strategy mechanism*, a mechanism (of the general sort described in the previous subsection) in which each participant, for each possible value of her private information, has one course of action that dominates all others, no matter what other participants in the mechanism do. Suppose the government, in our example, thought that it would like to see what its expected costs would be if it insisted on the "safety" of a dominant strategy mechanism for the two firms. How could it proceed to conduct an analysis?

Of course, we already know the bottom-line answer for the problem we have been analyzing. The government can construct a direct revelation mechanism, the nice mechanism, in which truth-telling is a dominant strategy for each firm and the expected costs to the government are 150. In this case there is no loss from insisting on a dominant strategy mechanism. But if we hadn't stumbled upon the nice mechanism, could we have come to this conclusion analytically?

We could have done so, in two steps. First, we can consider direct revelation mechanisms in which truth-telling (and participation) is a dominant strategy for both firms. We ask, For which direct revelation mechanisms (x_{mn}^i, t_{mn}^i) is participation and truth-telling a dominant strategy?

18.2. Optimal contracts for interacting parties

There are more constraints than before, but all the constraints are simple. For firm 1, we have four participation and four incentive constraints. The participation constraints are: For each of the four pairs of its actual cost and the cost reported by its rival, it must not sustain a loss by reporting its true cost. For example,

$$t_{21}^1 - 2x_{21}^1 \geq 0$$

is the constraint that firm 1 must not sustain a loss when it truthfully reports its costs are 2 and firm 2 (truthfully or not) reports costs of 1. The incentive constraints are: For each of the four pairs of its actual cost and the cost reported by its rival, the firm must prefer truth-telling to lying about its own cost. So, for example,

$$t_{21}^1 - 2x_{21}^1 \geq t_{11}^1 - 2x_{11}^1$$

expresses the constraint that firm 1, when its costs are 2 (note the cost premultiplying the production quantity it is allocated) and when its rival reports costs of 1, prefers to report truthfully that its costs are 2 rather than misrepresenting its costs as 1. If we satisfy all eight of these constraints for firm 1 and all eight for firm 2, we have a direct revelation mechanism in which participation and truth-telling are dominant.^h (We are only requiring weak dominance. To be really safe, the government might wish for strict dominance, in which case we could sharpen the constraints by adding a few pennies difference in the right direction in each.)

The government, then, can solve the problem of minimizing its expected costs of

$$\frac{1}{4}[t_{11}^1 + t_{11}^2] + \frac{1}{4}[t_{21}^1 + t_{21}^2] + \frac{1}{4}[t_{12}^1 + t_{12}^2] + \frac{1}{4}[t_{22}^1 + t_{22}^2]$$

subject to the constraints above that ensure that the mechanism has truth-telling as a dominant strategy. (We impose any appropriate nonnegativity constraints, and we insist on $x_{mn}^1 + x_{mn}^2 \geq 100$, so the government is sure to get its hundred units.) Once again we have a linear programming problem, and we can go to the computer to find a solution. When we did this, the computer returned a minimum cost to the government of 150 (which, given

^h Compare the participation constraints with those we imposed when we considered cases in which the firms could opt out of the mechanism ex post. In a direct revelation mechanism, where ex post opting out means opting out after all the players' types become common knowledge, the two are the same.

that we know about the nice mechanism, is hardly a surprise), and it gave the mechanism: $x_{11}^2 = x_{12}^2 = x_{22}^1 = x_{12}^1 = 100$, $t_{12}^1 = t_{22}^1 = 200$, $t_{11}^2 = t_{21}^2 = 100$, and everything else equal to zero. In words, firm 2 makes all the hundred units if it reports costs of 1, for which it is paid 100. If firm 2 reports costs of 2, firm 1 is given the assignment to make the hundred units and is paid 200. (If it isn't evident, go through the steps to show that this mechanism does indeed have truth-telling and participation as a dominant strategy for each firm.)

Can the government do better with some more complex dominant strategy mechanism? Of course, since we already know that 150 is the best we can do in any Nash equilibrium for any complex mechanism, we know that the answer is no. But for problems where the answer isn't apparent, we give the following general result.

The revelation principle for dominant strategy mechanisms. The outcome of any dominant strategy mechanism can be achieved in a direct revelation mechanism for which truth-telling and participation is a dominant strategy.

The proof is left as an exercise. If you understood the proof of our first revelation principle, this one should be easy.

General discussion

We have couched our discussion in terms of our simple toy problem, but it should be apparent that what we have said about direct revelation mechanisms, the revelation principle, and the revelation principle for dominant strategy mechanisms generalizes considerably.

The problem we have tackled can be stated roughly as: *Which mechanism among all possible mechanisms is optimal for achieving some given end?* Again roughly, we tackled this question in a two-step attack. First, we restricted attention to "direct revelation mechanisms," and specifically to truth-telling behavior in direct revelation mechanisms. Then via a revelation principle, we were able to show that restricting attention from "all possible mechanisms" to "direct revelation mechanisms" was without loss of generality. Consider each of these two steps in turn.

The problem of finding an optimal direct-revelation mechanism in which truth-telling is a Nash equilibrium or is a dominant strategy equilibrium is a fairly simple mathematical programming problem, because the conditions that "truth-telling is Nash" and "truth-telling is dominant" are expressible as a set of inequalities. This continues to hold in problems

more general than our toy problem, although our toy problem was special in two ways that made the problem we looked at especially simple:

- (1) In our toy problem, each of the firms had only a finite number of types (i.e., costs). Many applications in the literature deal with a continuum of types (e.g., the firms' levels of costs can be any number within some closed interval); and one will have to find ways of analyzing the problem algebraically instead of numerically.
- (2) The linearity of the cost structures in our toy problem meant that all the constraints were linear. If instead the firms had nonlinear cost functions, we would be doubly cursed. For one thing, we couldn't apply linear programming. For another, even if the cost function is convex, if we wrote the incentive constraints, we would be putting a convex function on both sides of the inequality, which means that, viewed as an optimization problem, what we would have would not necessarily be well behaved.ⁱ

In the toy problem, we discovered that it costs the government no more to have a mechanism in which truth-telling is dominant than to have one in which truth-telling constitutes a Nash equilibrium. You might wonder whether this is a fluke or the tip of some general result. It is, so to speak, both. If the firms have three possible costs and if those costs are not independently distributed, then it can be costly for the government to insist on dominant strategy implementation. (If you have access to a linear programming package, you can verify this in problem 8.) Hence the result we obtained is somewhat special. On the other hand, there are general classes of problems for which dominant strategy implementation costs no more than Nash implementation; this is so, for example, for a range of optimal auction design problems (see Bulow and Roberts, forthcoming.) A general analysis of this question is given by Mookherjee and Reichelstein (1989b).

Now we turn to the second part of our general attack: the demonstration that we can restrict attention to truth-telling in a direct revelation game, to see which outcomes can be "implemented" in general mechanisms.

The term *outcome* as used here in general means the selection of some state of affairs as a function of the private information held by parties for

ⁱ The reader may wish to consider whether, in the spirit of the analysis of chapter 16, we could employ some substitution of variables to make the constraints linear, even if the objective function is nonlinear.

whom the mechanism is to be designed. An outcome in this sense is something more than just the expected costs to the government; an outcome specifies how much each firm produces and what transfer is made to the firm as a function of the firms' costs. In our toy problem, and in many problems of *optimal* mechanism design, all that matters to the mechanism designer about the implemented outcome is some one-dimensional measure of goodness, such as expected cost. But in other applications, such as the one in the next section, we would want to look at the implementability of outcomes in the more general sense.

The term "implementable" is put in quotes because it can take on a number of different meanings, such as:

- (1) There is some mechanism with a Nash equilibrium (in the associated game of incomplete information) which gives this outcome.
- (2) There is some mechanism for which all the Nash equilibria give this outcome.
- (3) There is a mechanism for which there is a unique Nash equilibrium, and this Nash equilibrium gives this outcome.
- (4) There is a mechanism in which the outcome arises from the play of dominant strategies.

Each of these is progressively stronger,^j and each is presumably progressively more desirable for the mechanism designer, in order to have faith in the participants finding their way to the desired outcome. To repeat what we said before, if a mechanism admits several Nash equilibria, some of which are worse for the designer (and, more importantly, better for the participants) than is the equilibrium that is desired, then one worries that the participants will find their way to the wrong equilibrium. And even if the equilibrium is unique, one might worry whether the participants have the sophistication needed to find the equilibrium strategies (or that they have sufficient faith in their fellow participants' sophistication, so that they are happy to carry out their part of the equilibrium). If the desired outcome is the product of dominant strategy choices by the participants, then the mechanism designer can (presumably) have the greatest faith in the mechanism she has designed.

These four possibilities are far from exhaustive. You can probably think of other variations on this general theme, and you will find most of those variations somewhere in the literature.

^j To be precise, in (4) we must worry about other equilibria in weakly dominated strategies, so (4) isn't quite stronger than (3).

We have seen two revelation principles, which claim that anything that can be implemented in the sense of (1) or (4) can be implemented in the corresponding sense in the form of truth-telling in direct revelation games. These results are true in substantial generality.

We have not seen revelation principles corresponding to (2) and (3) as the notion of implementation. In general, the correspondence between truth-telling in direct revelation games and outcomes of general mechanisms breaks down when the solution concept is adapted from (2) or (3). But it is possible to augment a direct-revelation game, adding signals that the participants send (beyond a declaration of type), and then to obtain "augmented revelation principles" that speak to implementation in the sense of (2) and (3). That is, for any outcome that is implementable in the sense of, say, (2) in a general mechanism, there is an augmented direct revelation mechanism for which truth-telling is a Nash equilibrium that gives this outcome, and all other Nash equilibria give this outcome. The observation that this is possible is made in Ma, Moore, and Turnbull (1988), and the corresponding general "augmented revelation principles" are given by Mookherjee and Reichelstein (1989a).

Some outcomes cannot be implemented in the sense of (2), but there are mechanisms such that every subgame perfect Nash equilibrium for the mechanism gives this outcome, even though some subgame imperfect equilibria give other outcomes. And for outcomes that cannot be implemented in the sense of (2), there are mechanisms such that all the Nash equilibria that involve the play of no dominated strategy give this outcome, even though other Nash equilibria involving dominated strategies give other outcomes. That is, we can implement more (in the rough sense of [2]) if we look not at Nash equilibria but at some suitable refinement of Nash. For developments along these lines, see Moore and Repullo (1988) and Palfrey and Srivastava (1987).

All of the extensions and elaborations on optimal contract design that were described in section 18.1 can be undertaken in this more general and complex context. In particular, mixtures of adverse selection and moral hazard are possible. We might imagine, for example, that our two firms' costs are not determined entirely endogenously but result as well from R&D and investment decisions made by the firms. Then the government, besides buying its hundred units, might seek to put in place a mechanism that induces firms to invest optimally in plant and in R&D, where "optimality" means "that make the government's costs as low as possible." In principle, one can use the revelation principle to tackle such problems; in practice, they become rather complex. See Laffont and Tirole (1987) for an example.

18.3. The pivot mechanism

In the literature are a number of applications of the general ideas just given to the design of optimal mechanisms that deserve your attention. In particular, the analysis of optimal auction design is especially rich and well developed; references will be supplied at the end of this chapter.

Rather than pursue one of these, we look at a different sort of problem in mechanism design that uses some of the notions of the previous section. Instead of looking for an “optimal” mechanism, according to the interests of some individual who (presumably) is designing the mechanism with her own interests in mind, we will look at a classic example of mechanism design where the objective is to satisfy a set of criteria.

Consider the following story. Several farmers live on the banks of a stream. (Some live on one side, some on the other.) The number of farmers is I , and we index them by $i = 1, 2, \dots, I$. They consider building a bridge over the stream that will permit them all to cross back and forth. There is only one place where the bridge can be constructed, and the bridge will cost K to construct. Each farmer attaches some value to having the bridge constructed, but none is quite sure what value the others attach to it. If they construct the bridge, the funds to do so will have to come from their own pockets.”

These farmers must decide whether or not to build this bridge. We take it as given that if the bridge is built, each farmer will pay K/I , his pro rata share of the costs of building the bridge. But in addition to this, the farmers are willing to consider transfers among themselves; we write t^i for the amount of money taken from farmer i , above and beyond the K/I that is collected if the bridge is to be built. We do not preclude $t^i < 0$, which means that a subsidy is paid to farmer i .

Farmer i 's utility depends on whether the bridge is built and on any monetary transfer t^i that is made to or from him. We suppose that each farmer i attaches some monetary value to the bridge, u^i , and that farmer i 's utility is linear in money and in this valuation. That is, farmer i 's utility depends on (a) whether the bridge is built, (b) the building tax K/I that he pays if it is built, and (c) any further tax or subsidy, t^i ; this utility is given by

$$\text{Farmer } i\text{'s utility} = \begin{cases} u^i - K/I - t^i & \text{if the bridge is built, and} \\ -t^i & \text{if the bridge isn't built.} \end{cases}$$

Note that we permit transfers even if the bridge isn't built. We assume that farmers may dislike having this bridge built; i.e., $u^i < 0$ is possible.

18.3. The pivot mechanism

In fact, we assume that u^i could be any real number whatsoever; no value is precluded.

It will be expositionally convenient to work with the farmer's valuation of the bridge, *net* of his contribution for building it, or $v^i = u^i - K/I$. Note that with this substitution, farmer i 's utility is $v^i - t^i$ if the bridge is built.

The farmers meet together one evening to try to decide whether to build the bridge and what taxes and subsidies to enforce beyond the pro rata contribution to building it. One farmer suggests majority rule and no transfers: Put building the bridge to a vote, and if a majority vote to build the bridge, then everyone pays only his contribution K/I ; if a majority vote against, no transfers are made at all. If this plan is adopted, farmers will vote for the bridge if and only if their net personal valuation v^i exceeds 0. But there are several objections to this. One farmer observes that some farmers may really want this bridge, and if most farmers don't care that much (i.e., if $u^i < K/I$ for many farmers), then the bridge won't be built, even though general social welfare would be improved if it were built. Another objects that he doesn't want this bridge much at all, in fact he positively dislikes it, and yet he may be assessed with a building fee of K/I . What justice is there in that?

So someone suggests that farmers “subscribe” to the bridge. Everyone will write his name and a pledge (an amount of money) on a piece of paper and toss it into a hat. After all the pledges are collected, they will be added together. Negative pledges are allowed; farmers in this way demand compensation for having the bridge built. If the sum of the pledges exceeds 0, the bridge will be built, and every farmer must contribute the amount of his pledge (plus his share K/I). If the pledges amount to less than 0, then the bridge isn't built and no transfers will be made. What to do with the surplus, if more than zero is pledged? One farmer suggests that this surplus be divided equally among all the farmers. Another suggests that it be divided proportionally (in proportion to original pledges) among those farmers who pledged positive amounts. Still another suggests that it might be a good thing to collect the excess and burn it. (This last suggestion is met with some incredulity.)

Thinking about this mechanism, the farmers recognize some difficulties. Imagine a farmer who values the bridge at a net amount v^i . How much should he pledge? If he believes that it will take a pledge of v^i or more to get the bridge built, then he is willing to pledge precisely v^i . He certainly would never wish to pledge more than this, since he will be stuck with a pledge that is greater than the net value of the bridge to him. On the other hand, if he believes that a pledge of less than v^i will do, then he

wishes to shade his pledge down from v^i and "free-ride" on the pledges of his fellow farmers. In general, he won't be sure which of these two circumstances pertains, but as long as he assesses positive probability for the second, his optimal bid will be something less than v^i . Hence the total pledges will be less than $\sum_i v^i$, and the bridge may not be built when it should be. In any case, another farmer objects that figuring out how to behave in this mechanism is too hard for simple country bumpkins, and all sorts of bad outcomes may result.

After much discussion, the farmers decide that they wish to find some mechanism for deciding whether to build the bridge and what taxes to assess that has the following properties:

- (a) The bridge will be built if and only if it is socially efficient to do so; that is, if and only if $\sum_i v^i \geq 0$.
- (b) No farmer should have to waste his time doing complicated analysis of how to behave in the mechanism. In particular, no farmer should have to spend time trying to assess what his fellow farmers will do. Put another way, the optimal actions of each farmer in the mechanism (as a function of the farmer's private valuation v^i) should dominate any other actions the farmer might take, no matter what his fellow farmers do.
- (c) The mechanism, if played optimally by a farmer, should never be so injurious to the welfare of the farmer that he would prefer that the decision to build or not to build is taken by decree. That is, farmer i should not wind up with utility less than $\min\{v^i, 0\}$.
- (d) Since no outsider is willing to put up money to permit the mechanism to function, the taxes collected (less any subsidies, and not including the building tax K/I per farmer, if the bridge is built) must be nonnegative.

One can certainly quibble (a), (b), and (c) (and later we will have a very large quibble with [a], in particular), but let us accept them and see where they lead.

We translate (b) as saying that the farmers want a dominant strategy mechanism. They wish to design a general mechanism in which each farmer, as a function of his personal valuation v^i , has a dominant strategy to play. With this, we appeal to the revelation principle for dominant strategy mechanisms: Anything we can do with a general dominant strategy mechanism can be done with a direct revelation mechanism in which truth-telling is a dominant strategy. So we can restrict attention to direct revelation mechanisms.

In this context a direct revelation mechanism takes the form: Each farmer is asked to reveal his personal valuation v^i . We put hats on variables to distinguish what farmers reveal from their true valuations; that is, $\hat{v}^i$ is what farmer i reveals. As a function of the vector of revealed valuations $\hat{v} = (\hat{v}^1, \dots, \hat{v}^I)$, a decision is made whether to build the bridge or not, and taxes on each farmer are determined. We write $\alpha(\hat{v})$ for the decision whether to build the bridge, where $\alpha(\hat{v}) = 1$ means that the bridge is built and $\alpha(\hat{v}) = 0$ means it is not. (In a more general treatment of these issues, we might let $\alpha(\hat{v})$ take on any value in the interval $[0, 1]$, interpreting this as the probability that the bridge will be built. But given the farmers' requirement [a], we know what α must be, and this doesn't entail randomized decisions whether to build the bridge.) And we write $t^i(\hat{v})$ for the tax imposed on farmer i . (More generally, we could permit random taxes, but as farmers are risk neutral, this would add nothing to the story except horrid notation.)

It will be handy to have the following pieces of notation. We let $\hat{v}^{-i} = (\hat{v}^1, \dots, \hat{v}^{i-1}, \hat{v}^{i+1}, \dots, \hat{v}^I)$. That is, $\hat{v}^{-i}$ is the vector of reported valuations for all the farmers except for i . We sometimes write $t^i(\hat{v})$ as $t^i(\hat{v}^i, \hat{v}^{-i})$; i.e., we put i 's valuation as the first argument. Finally, we write $\hat{\Sigma}$ for $\sum_{j=1}^I \hat{v}^j$ and $\hat{\Sigma}^{-i}$ for $\sum_{j \neq i} \hat{v}^j$. That is, $\hat{\Sigma}$ is the sum of all the announced valuations, and $\hat{\Sigma}^{-i}$ is the sum of all the announced valuations save i 's.

We are looking for a direct revelation mechanism in which truth-telling is a dominant strategy and in which the bridge is built if and only if $\sum_i v^i \geq 0$. The latter restriction, combined with the notion that truth-telling will be dominant tells us what the function α must be:

$$\alpha(\hat{v}) = \begin{cases} 1, & \text{if } \hat{\Sigma} \geq 0, \text{ and} \\ 0, & \text{if } \hat{\Sigma} < 0. \end{cases} \quad (1)$$

Furthermore, we have the following results:

Lemma 1. *The taxes paid by farmer i must take the form*

$$t^i(\hat{v}^i, \hat{v}^{-i}) = \begin{cases} \bar{t}^i(\hat{v}^{-i}), & \text{if } \hat{\Sigma} \geq 0, \text{ and} \\ \underline{t}^i(\hat{v}^{-i}), & \text{if } \hat{\Sigma} < 0, \end{cases} \quad (2)$$

for $\bar{t}^i$ and $\underline{t}^i$ functions of $\hat{v}^{-i}$.

The idea is that what a farmer pays in taxes cannot depend on what he himself reveals as his valuation, except insofar as this revelation changes the decision whether or not to build the bridge. To see why this is so,

suppose v^i and w^i , two valuations for i , and $\hat{v}^{-i}$ are such that $v^i + \hat{\Sigma}^{-i} > 0$, $w^i + \hat{\Sigma}^{-i} > 0$, and $t^i(v^i, \hat{v}^{-i}) > t^i(w^i, \hat{v}^{-i})$. In words, if the other farmers are announcing $\hat{v}^{-i}$, then whether i announces v^i or w^i , the bridge will be built. But announcing v^i (when the others are announcing $\hat{v}^{-i}$) results in higher taxes than does announcing w^i . In such circumstances, the mechanism doesn't have truth-telling as a dominant strategy; farmer i , if his valuation is v^i , would prefer to misrepresent his valuation as w^i (if his fellow farmers announce $\hat{v}^{-i}$). This establishes the first part of the proposition: For $\hat{v}$ such that $\hat{\Sigma} \geq 0$, $t^i(\hat{v})$ depends only on $\hat{v}^{-i}$. A similar argument gives the other half.

Lemma 2.

$$\bar{t}^i(\hat{v}^{-i}) - \underline{t}^i(\hat{v}^{-i}) = -\hat{\Sigma}^{-i}. \quad (3)$$

To prove this, fix $\hat{v}^{-i}$ and consider the case where $v^i = -\hat{\Sigma}^{-i}$. If farmer i truthfully reveals v^i when the others reveal $\hat{v}^{-i}$, then farmer i must prefer revealing v^i to revealing $v^i - \epsilon$ for $\epsilon > 0$. But, in these circumstances, revealing v^i means the bridge will be built and revealing $v^i - \epsilon$ means that the bridge won't be built. So by revealing v^i , farmer i nets $v^i - \bar{t}^i(\hat{v}^{-i})$, while revealing $v^i - \epsilon$ nets $-\underline{t}^i(\hat{v}^{-i})$. The former must be at least as large as the latter, so

$$v^i = -\hat{\Sigma}^{-i} \geq \bar{t}^i(\hat{v}^{-i}) - \underline{t}^i(\hat{v}^{-i}).$$

Now consider the case where $v^i = -\epsilon - \hat{\Sigma}^{-i}$ for $\epsilon > 0$. Truthful revelation of v^i causes the bridge not to be built, giving farmer i utility $-\underline{t}^i(\hat{v}^{-i})$. Falsely revealing $v^i + \epsilon$ causes the bridge to be built, and farmer i has utility $v^i - \bar{t}^i(\hat{v}^{-i})$. Since the former must be at least as large as the latter (to support truth-telling),

$$v^i = -\epsilon - \hat{\Sigma}^{-i} \leq \bar{t}^i(\hat{v}^{-i}) - \underline{t}^i(\hat{v}^{-i}).$$

This is true for arbitrary $\epsilon > 0$, so by letting $\epsilon \rightarrow 0$,

$$-\hat{\Sigma}^{-i} \leq \bar{t}^i(\hat{v}^{-i}) - \underline{t}^i(\hat{v}^{-i}).$$

Together with the next to last inequality, this gives the desired result.

Lemma 3.

$$\underline{t}^i(\hat{v}^{-i}) \leq \begin{cases} \hat{\Sigma}^{-i}, & \text{if } \hat{\Sigma}^{-i} \geq 0, \text{ and} \\ 0, & \text{if } \hat{\Sigma}^{-i} < 0. \end{cases} \quad (4)$$

Consider first the case where $\hat{\Sigma}^{-i} \geq 0$. Suppose $v^i = 0$. By truthfully reporting v^i , farmer i causes the bridge to be built. His utility in this case is $v^i - \bar{t}^i(\hat{v}^{-i}) = 0 - [\underline{t}^i(\hat{v}^{-i}) - \hat{\Sigma}^{-i}]$. Since (c)⁹ entails that farmer i is no worse off for having revealed the truth than 0 (in utility), we have $-\underline{t}^i(\hat{v}^{-i}) - \hat{\Sigma}^{-i} \geq 0$ or $\underline{t}^i(\hat{v}^{-i}) \leq \hat{\Sigma}^{-i}$. Consider next the case where $\hat{\Sigma}^{-i} < 0$. Again suppose $v^i = 0$. By truthfully revealing v^i , farmer i causes the bridge not to be built. Hence he nets $-\underline{t}^i(\hat{v}^{-i})$. Since he must be no worse off than 0 by so doing, we have $-\underline{t}^i(\hat{v}^{-i}) \geq 0$, or $\underline{t}^i(\hat{v}^{-i}) \leq 0$.

Lemma 4.

$$\underline{t}^i(\hat{v}^{-i}) = \begin{cases} \hat{\Sigma}^{-i}, & \text{if } \hat{\Sigma}^{-i} \geq 0, \text{ and} \\ 0, & \text{if } \hat{\Sigma}^{-i} < 0. \end{cases} \quad (5)$$

In other words, inequality (4) must be an equation.

To see this, we note first that from lemmas 1 and 2, (d) rendered in symbols is

$$\sum_i \underline{t}^i(\hat{v}^{-i}) \geq 0 \text{ if } \hat{\Sigma} < 0, \text{ and} \quad (d1)$$

$$\sum_{n=1}^N \bar{t}^i(\hat{v}^{-i}) = \sum_i [\underline{t}^i(\hat{v}^{-i}) - \hat{\Sigma}^{-i}] \geq 0 \text{ if } \hat{\Sigma} \geq 0. \quad (d2)$$

Suppose, then, that for some i and $\hat{v}^{-i}$ such that $\hat{\Sigma}^{-i} \geq 0$ we had a strict inequality in (4). That is, $\underline{t}^i(\hat{v}^{-i}) < \hat{\Sigma}^{-i}$. Choose v^i to be sufficiently large so that $\hat{\Sigma}^{-i'} > 0$ for all i' and so that $\hat{\Sigma} > 0$. The bridge will be built, and by lemmas 1 and 2, farmer i' pays

$$\bar{t}^{i'}(\hat{v}^{-i'}) = \underline{t}^{i'}(\hat{v}^{-i'}) - \hat{\Sigma}^{-i'}.$$

By lemma 3, each such term is nonpositive and, by the supposition, the term for i is strictly negative. Hence the sum of all the payments is strictly negative, contradicting (d2).

Similarly, suppose that for i and $\hat{v}^{-i}$ such that $\hat{\Sigma}^{-i} < 0$, we had $\underline{t}^i(\hat{v}^{-i}) < 0$. We can choose v^i sufficiently small so that $\hat{\Sigma}^{-i'} < 0$ for all i'

⁹ In case you forgot, (c) says that, in general, a farmer should do no worse than $\min\{v^i, 0\}$ by reporting the truth.

and $\hat{\Sigma} < 0$. Arguing along the lines just given, we obtain a violation of (d1).

We now put all the pieces together and provide a converse.

Proposition 2. *There is only one direct revelation mechanism for which (a) through (d) hold, namely the direct revelation mechanism defined by (1), (2), (3), and (5). This mechanism is alternatively defined by (1) and*

$$t^i(\hat{v}) = \begin{cases} 0, & \text{if } \hat{\Sigma} \geq 0 \text{ and } \hat{\Sigma}^{-i} \geq 0, \\ 0, & \text{if } \hat{\Sigma} < 0 \text{ and } \hat{\Sigma}^{-i} < 0, \\ \hat{\Sigma}^{-i}, & \text{if } \hat{\Sigma} < 0 \text{ and } \hat{\Sigma}^{-i} \geq 0, \text{ and} \\ -\hat{\Sigma}^{-i}, & \text{if } \hat{\Sigma} \geq 0 \text{ and } \hat{\Sigma}^{-i} < 0. \end{cases} \quad (6)$$

Proof. The lemmas establish that this is the only possible candidate. Showing that (6) is equivalent to (2), (3), and (5) is a matter of simple bookkeeping. We see from (6) that $t^i(v) \geq 0$ in all cases, so (d) is obvious. So all that remains is to show that the mechanism defined by (1) and (6) satisfies (b) and (c). We will do half of this, leaving the other half for homework.

Suppose that, for some i and $\hat{v}^{-i}$, $\hat{\Sigma}^{-i} \geq 0$ and $v^i + \hat{\Sigma}^{-i} \geq 0$. If i tells the truth, the mechanism calls for the bridge to be built and for i to pay no tax. The only way that i can change the outcome is to report (falsely) a valuation $\hat{v}^i$ less than $-\hat{\Sigma}^{-i}$. Then the bridge won't be built, and i will pay a tax of $\hat{\Sigma}^{-i}$. Reporting truthfully, then, leaves i with utility v^i , and a misrepresentation sufficient to change the outcome leaves i with utility equal to $-\hat{\Sigma}^{-i}$. Since we assume in this case that $v^i + \hat{\Sigma}^{-i} \geq 0$, $v^i \geq -\hat{\Sigma}^{-i}$. Thus, telling the truth is a best reply; also, i ends up with utility v^i , so (c) holds in this case.

Suppose that, for some i and $\hat{v}^{-i}$, $\hat{\Sigma}^{-i} < 0$ and $v^i + \hat{\Sigma}^{-i} \geq 0$. The bridge will be built if i reports truthfully, and he will pay a tax of $-\hat{\Sigma}^{-i}$, which gives i utility $v^i + \hat{\Sigma}^{-i} \geq 0$. So (c) holds. Moreover, the only way that i can change the outcome (and his utility) is by reporting a valuation $\hat{v}^i < -\hat{\Sigma}^{-i}$. But if he does this, $\hat{\Sigma}^{-i} < 0$ and $\hat{\Sigma} < 0$, which means that there is no transfer, and i winds up with utility 0. So misrepresenting is not beneficial.

(The other two cases are left for you.)

This particular mechanism is called the *pivot mechanism* because a tax is paid by farmer i only if his valuation v^i is pivotal, that is, only if his valuation changes the decision from what it would be if he reported zero.

Moreover, when i 's valuation is pivotal, i is taxed an amount that is just equal to the "social distress" his pivotal valuation causes; i.e., if he causes the bridge to be built when it otherwise wouldn't be, he pays $-\hat{\Sigma}^{-i}$, which is the monetary "cost" to the rest of the farmers of the bridge; and if he causes the bridge not to be built when it otherwise would be, he pays $\hat{\Sigma}^{-i}$, the monetary benefit to the other farmers of the bridge.

This mechanism satisfies our four criteria, and in fact it is the only mechanism that does this, so if we accept the four as desirable, we have come upon quite a nice result. Condition (a), however, seems especially suspect. The justification for it is that we wish to achieve a "social optimum," which in this society where utility is linear in money means maximizing the sum of individual utilities. But the mechanism doesn't achieve a social optimum if there are any pivotal individuals, because it produces a positive net collection of taxes. Note well: The transfers are all nonnegative, and they are all zero only if there are no pivotal individuals.

This raises the question, If there are pivotal individuals, so a net surplus of funds is collected, what happens with that surplus? Unthoughtful answers are to give it back to the farmers, or to hold it for the next project to come along, or to use it to throw a dance. If the surplus is used in any fashion that gives utility to the farmers, and if the farmers anticipate this, then the direct revelation mechanism isn't what we described. We would have to include in the farmer's utilities the value they attach to the uses to which the surplus is put. But our uniqueness result is that the only mechanism that will achieve the four required properties is the one described. This means that if we wish to achieve (a) through (d), we must find a use for the surplus that is of no benefit (or detriment) to our farmers. The surplus must be burned, or mailed off to aid some other lucky group of individuals about whom none of our farmers cares in the least. (And our farmers can't anticipate that any such largesse may result in a reciprocal gift coming to them. It is probably safest to burn any surplus.)

Condition (a), then, can be attacked on the grounds that it doesn't guarantee that a social optimum is reached. The decision whether to build the bridge will be done "optimally," but at a waste of other social resources, if there are any pivotal individuals.

We can put this another way. Condition (a) would imply that a social optimum is reached if it were joined to the following modification of (d): The sum of taxes and transfers, not including the taxes collected to build the bridge if the bridge is built, totals zero precisely. This condition is known in the literature as the *balanced budget condition*, and what we know from our analysis is that it is impossible to satisfy this together with (a), (b), and (c). In fact, one can show that the balanced budget

condition is inconsistent with (a) and (b) alone. It is impossible to achieve a Pareto optimal outcome in this closed society with a dominant strategy mechanism.

18.4. The Gibbard-Satterthwaite theorem

One way to view the results of the previous section is that asking for a dominant strategy mechanism may be asking too much. We close by alerting you to another result that reinforces this message (or which, at least, implies that asking for a dominant strategy mechanism is asking for rather a lot).

Recall Arrow's possibility theorem from chapter 5. The setting is one with a finite number I of individuals, and a finite number of possible social outcomes. We let X be the set of social outcomes. Each individual has preferences $\succ_i$ defined on X that are asymmetric and negatively transitive.

In Arrow's theorem, we were concerned with aggregation of the array of society's preferences $(\succ_1, \dots, \succ_I)$ into a social ordering $\succ^*$ on X . The story before was that members of society attempted to design some social choice rule that would take the array of preferences $(\succ_1, \dots, \succ_I)$ and transform them in desirable fashion into society's preferences. We saw that a few, seemingly quite desirable properties for the social choice rule, forces social choice to be dictatorial, which doesn't seem very desirable.

One thing unexplained was how, even if we had found a nice social choice rule, we were going to work out what the preferences of each member of society were. Perhaps the preferences of individual i would be obvious. But it seems more likely that at some point individual i would have to volunteer what she liked and didn't like. And even if we had a nice social choice rule, we might worry that individual i would misrepresent her preferences in an attempt to get social preferences more to her liking.

So we can approach the matter as one of mechanism design. Suppose we had a desirable social choice rule. In fact, we'll make things easier on ourselves: Suppose we had a desirable social choice function ϕ that associates with every array of individual preferences $(\succ_1, \dots, \succ_I)$ some outcome $\phi(\succ_1, \dots, \succ_I) = x \in X$, which is the outcome that is implemented. All we want to know is whether the given ϕ can be implemented "reliably" by some mechanism, where what we mean by reliable implementation is that if individual preferences are given by $(\succ_1, \dots, \succ_I)$, then the mechanism (which involves actions of some sort by the members of society, each of whom knows her own preferences) has *dominant strategies* for members of society that lead to the outcome $\phi(\succ_1, \dots, \succ_I)$. By the revelation principle for dominant strategy mechanisms, we can turn this into the following question:

Given ϕ , is there a direct revelation mechanism, where each individual is called upon to reveal her preferences over X , which has truth-telling as a dominant strategy and which results in the outcome $\phi(\succ_1, \dots, \succ_I)$ when preferences are $(\succ_1, \dots, \succ_I)$?

The Gibbard-Satterthwaite theorem. If the domain of ϕ is the space of all I -tuples of preferences over X , if the range of ϕ has at least three elements, and if ϕ can be implemented by a dominant strategy mechanism, then ϕ must be dictatorial; there is some i such that $\phi(\succ_1, \dots, \succ_I)$ is one of the $\succ_i$ -most preferred elements of X .

For elegant proofs, see Schmeidler and Sonnenschein (1978) and Barbera (1983).

You may find the conjunction of the Gibbard-Satterthwaite theorem and proposition 2 (on the pivot mechanism) a bit jarring. The pivot mechanism implements a particular social choice function with dominant strategy mechanism, yet that social choice function is hardly dictatorial. As we noted in the previous section, the social choice function of the pivot mechanism is not efficient. But the Gibbard-Satterthwaite theorem says nothing about efficiency of the social choice function. So how are these reconciled?

They reconcile because the Gibbard-Satterthwaite theorem requires that domain of the social choice function is the set of *all* possible I -tuples of preferences over the social outcome. In the farmers and bridge story, preferences over social outcomes come from a very restrictive domain. Preferences are "quasi-linear" in the decision whether to build the bridge and the transfer received by a farmer, and farmers don't care at all about the transfers their neighbors receive. If we allowed farmers to have all manner of preferences over the full social outcome, proposition 2 would crumble, as the Gibbard-Satterthwaite theorem says it must.

18.5. Bibliographic notes

The general subject of optimal contract and mechanism design covers many different categories in the literature. In this chapter, we have limited attention to contract design when adverse selection (or hidden information) is the issue; compare with chapter 16, where we studied some basic results in optimal contract design in situations of moral hazard. (Although we didn't discuss it except in passing in chapter 16, issues related to implementation of particular sets of actions for interacting agents arises there, just as in this chapter. If you did problem 3 in chapter 16, you at least got a taste of this.) And while we have looked at problems that involve only moral hazard or only adverse selection, problems that mix the two can be tackled by a mixture of the methods we have explored.

At the same time, the techniques discussed in this chapter are used to study optimal contracts and problems in social choice, where the objective is to see which social choice rules can be implemented in particular environments. (Sections 18.3 and 18.4 gave two examples of the latter sort of activity.) Finally, there is a wide range of applications of these techniques.

Sending you out into the literature, therefore, is a formidable task. It is hard to know where to begin, and once a list is started, it is harder to know where to stop. So we make very limited suggestions.

(1) A neat application of optimal contract design for a single party is found in Baron and Myerson (1982), where they study how to regulate a monopolist whose costs are unknown to the regulatory body. We did nothing at all with the problem of mechanism design by an informed party; on this topic, Myerson (1983) is seminal.

(2) The issues tackled with the toy problem of section 18.2 are especially well developed in the domain of optimal auction design. Myerson (1981) is seminal, McAfee and McMillan (1987) and Milgrom (1987, 1989) give very nice, up-to-date surveys, and Bulow and Roberts (forthcoming) is recommended for the way in which they reduce auction design to a classical problem in price theory.

(3) For an analysis of a problem of optimal mechanism design that mixes adverse selection and moral hazard, see Laffont and Tirole (1987).

(4) On the general topics of sections 18.3 and 18.4, Green and Laffont (1979) is a good place to begin. The introduction to Moore and Repullo (1988) provides an excellent summary of recent developments concerning mechanism design in the context of general social choice.

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18.6. Problems

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18.6. Problems

- 1. Recall that in chapter 9, we considered a large manufacturing monopolist selling to individual retailing monopolists. Recall that in one parameterization of the problem, there were ten retailers, three of whom faced demand of the form $p = 12 - 9x$, five of whom faced demand of the

form $p = 12 - 6x$, and two who faced demand of the form $p = 12 - 2x$. Costs of production and sales were zero. The retailers are price-takers in their dealings with the manufacturing monopolist.

(a) Assume that the manufacturer can tailor a take-it-or-leave-it offer for each individual retailer, of any form desired. To maximize her profits, what should the manufacturer do?

(b) Suppose the manufacturer is compelled by law to make identical offers to all ten retailers, but she can otherwise make take-it-or-leave-it offers, offers with menus where retailers are restricted to taking only one item from the menu, and so on. What is the best the manufacturer can do in this instance?

■ 2. Consider the problem of designing an optimal insurance contract as discussed in section 18.1. Prove that it is never optimal to offer consumers a "random contract," where by a random contract we mean one in which the amount the consumer is left with, y_1 or y_2 , is random beyond the randomness caused by the initial endowment risk. (You should assume that any additional randomness has the same distribution for high- and low-risk consumers.) Compare this with problem 12 in chapter 3.

■ 3. In the optimal menu of insurance contracts offered to the consumer in section 18.1, if $\rho > 0$ is it ever optimal to pool the two types of consumer? If $\rho < 1$ is it ever optimal to offer only full insurance to high-risk consumers, leaving low-risk consumers uncovered?

■ 4. Suppose that the insurance company of section 18.1 was (for some reason) constrained to offer a single insurance policy. That is, menus are not permitted. What will the optimal (profit-maximizing) insurance policy look like? In particular, will it involve full insurance? (Hint: If you did problem 3, you should have a good start on this. If you need a bit of inspiration, go back to the analysis in chapter 9 of the manufacturer selling to retailers under the constraint that all the retailers had to be made the same single take-it-or-leave-it offer.)

■ 5. Suppose, in the context of the menu of insurance policies depicted in figure 18.6, a number of (cartelized) firms were offering the menu shown, and consumers could buy the policies from different firms simultaneously. We imagine the policies being in the form of a premium that is paid at the outset and, if the low-endowment state prevails, a payment that is made from the firm to the insuree. Hence buying two of the same contract would double both the premium and the contingent payments. (Don't

worry about the time value of money or the fact that the premium must be paid before the insuree's endowment is realized.) What would a high-risk consumer do in this case? What would a low-risk consumer do? How does this affect the profits of the insurance firms? (How would similar considerations affect the answers you gave to problem 5 in chapter 16?)

■ 6. Consider the toy problem of section 18.2. Imagine that the government institutes the following procedure: The two firms are asked to bid (simultaneously and independently) on the prices they will charge. They may not name a price greater than 4 or less than zero. (Of course, each knows its own unit cost when it does this.) If they name the same price, the order is split between them. If they name a different price, the order is given entirely to the firm naming the lower price. In both cases, the firm is paid per unit what it bids.

(a) If we construct from this mechanism the corresponding game (between the firms) of incomplete information, does the game have a pure strategy Nash equilibrium?

(b) If not, can you find a symmetric mixed strategy equilibrium?

(c) If you find any equilibrium, what is the corresponding direct revelation mechanism?

■ 7. Suppose we modify problem 6 in two ways. First suppose that the firms have costs of 1, 2, or 3, with each level of cost equally likely, and with the cost levels of the two firms independent of one another. (Each firm knows its own cost but not the costs of its rival.) Suppose the government solicits bids from the firms. If they bid the same amount, the order is split; if one firm bids less, it is given the entire order. And (the second change) a firm is paid not what it bid but what the other firm bid.

(a) If we construct from this mechanism the corresponding game of incomplete information, does this game have a pure strategy Nash equilibrium? Does it have more than one? If there is more than one, is there any reason to think that one equilibrium is more tenable than the others?

(b) For the equilibria you find in (a) construct the corresponding direct revelation games.

(This sort of auction, where the winning bidder is rewarded according to the bid made by the "second" best bid, is called a Vickrey auction. Almost any treatment of auction theory will tell you how to analyze this sort of auction, if you can't figure it out for yourself.)

■ 8. (Do this problem only if you have access to a linear programming package on a computer. Do not attempt to do it by hand.) Consider a problem similar to the toy problem of section 18.2. The government must procure one hundred units of some good. The good can be procured from two firms. Each firm has a linear cost technology with costs that are private information. Each firm's costs are either 1, 2, or 3 per unit produced. The marginal probabilities of each of the three costs prevailing at one of the firms are all $1/3$. But the costs are not statistically independent. The probability that both firms have costs of 1 is $1/6$, as is the probability that both have costs of 2 and the probability that both have costs of 3. The probability that firm 1 has costs of 1 and firm 2 has costs of 2 is $1/12$, as is the probability of all the other nonequal cost combinations.

If the government knew the costs of the two firms, it could get its hundred units at a cost of 100 with probability $1/2$, at a cost of 200 with probability $1/3$, and at a cost of 300 with probability $1/6$, for a total expected cost of 166.67. But, we suppose, the government doesn't know the costs of the two firms. In addition, we suppose that all that one firm knows about the costs of its rival is the value of its own costs. That is, firm 1 might know that its costs are 2, which means that it assesses probabilities $1/4$ that its rival's costs are 1, $1/2$ that its rival's costs are 2, and $1/4$ that its rival's costs are 3.

(a) Suppose the government seeks to find the best it can do (in terms of minimizing expected costs) in a Nash equilibrium of some mechanism. Firms must agree to participate after they learn their own costs, but then they are constrained to continue to participate. The government is not allowed to extract money from the firms; i.e., the transfers t^i must be nonnegative. What is the best the government can do?

(b) Suppose we remove from (a) the constraint that transfers must be nonnegative. What is the best the government can do? (If you find the answer striking, see Cremer and McLean [1988].)

(c) Suppose the government seeks to find the best it can do (in terms of minimizing expected costs in a dominant strategy mechanism). What is the minimum expected payment the government must make?

■ 9. Consider the mechanism depicted in figure 18.7(a). What are the Nash equilibria associated with this mechanism (that involve participation by both firms)? For one of those Nash equilibria, design the corresponding direct revelation mechanism and verify that participation and truth-telling is a Nash equilibrium in the corresponding direct revelation game.

■ 10. Finish the proof of proposition 2.

■ 11. In the context of the pivot mechanism, show that (a), (b), and the balanced budget condition are incompatible.

part V

Firms and transactions

chapter nineteen

Theories of the firm

In this last part of the book, we take a brief look at firms (and, implicitly, other sorts of economic institutions). Our purpose is two-fold. Needless to say, firms are important parts of most capitalist economies, and a better understanding of what they do and why they exist is important to any conception of the economy. At the same time, discussion of the current state of theorizing about firms gives us an excellent vehicle for exploring some of the weaknesses of the methods with which we have been concerned in the second half of this book and, in the end, with the methods that we have used throughout. In a sense, this part of the book and especially the postscript that follows serve as denouement. We take stock of the limits of the tools that formal microeconomic theory provides.

The plot is as follows: We begin in this chapter with models of the firm that have the firm as the unit of analysis when they become formal. The classic theory of the firm as a profit-maximizing production possibilities set (a wonderful subject for animation) is the starting point, and we begin by attacking the notion that profit maximization is what a firm does or ought to do. We then very briefly discuss so-called *managerial models of the firm*, where the firm is conceived of as a production possibilities set that doesn't (necessarily) maximize profits but pursues some other goal that reflects the interests of managers: maximization of sales revenue, or of capital, or of the probability of a "satisfactory" level of profits. Finally we consider a very different conception of the firm, due to Nelson and Winter (1982). In this approach, firms are dynamic entities, which at a given point of time, are described by a production *routine* and which evolve according to environmental factors through patterns of *search* and *imitation*, looking for better (but in no sense optimal) routines.

Although the firm is the unit of analysis in all these models, each is grounded in some finer conception of what goes on within the firm, and each is amenable to (some) finer-grained analysis. In chapter 20, we explore a general paradigm for finer-grained analysis, known as *transaction cost economics*, where the unit of analysis is the individual transaction. In this theory, firms are not entities, things of the rough category of the consumer; instead firms are institutions, in the rough category of the market.

Firms are places in which exchanges take place between individuals, exchanges that could take place in markets, but which are more efficiently consummated within a firm.

In this theory, the line between firms and markets is rather fuzzy, but what line there is is drawn along the dimension of the frequency of interaction, the relative permanence of certain legal and market relationships, and the extent to which parties to a transaction are "tied" to one another. The methods of part III and, in particular, chapter 14 seem natural for an explanation of such relationships. At the same time, firms both respond to and call up an array of informationally complex transactions, the stuff of part IV. So it seems natural to attack some of the basic notions of chapter 20 with those techniques as well. But the complexity of transactions within a firm and the inherent indeterminateness of equilibrium in situations of repeated interaction call into question both the standard models of individual dynamic choice and the relevance of equilibrium analysis, where every individual is well aware of the role she has to play. In the postscript, we will see that this presents both a weakness in the current state of formal theory and an opportunity for the development of new methods.

19.1. The firm as a profit-maximizing entity

In neoclassical theory, the firm is described by a production possibilities set, and it is assumed to choose the (feasible) production plan that maximizes profits. For the time being, we will accept the notion that those who control the firm know the production possibilities of the firm, and we will consider why the firm would seek to maximize its profits.

Some typically cited reasons are:

(1) *It is a good positive model of how firms act.*

This may well be the case, but if it is to be believed, we'll need either some empirical evidence or some more fundamental theoretical analysis that leads to this conclusion. An example of the latter might be:

(2) *Firms that profit maximize will drive out firms that don't by a process of natural selection.*

"Natural selection" of this sort might work, but just the bald assertion that this is so is hardly convincing. Some sort of detailed model of the natural selection process is needed. We will return to this in section 19.3.

19.1. The firm as a profit-maximizing entity

But for now we adopt a cynical demeanor and press on for other possible rationales for profit maximization.

(3) *If the managers of the firm don't maximize profits, then the firm will be taken over by some corporate raider (who can make money by maximizing profits), and the managers will be fired. Since managers don't want to be fired, they maximize profits.*

At least two things are missing from this argument. First, it requires that one have faith in the takeover mechanism; that is, we must believe (or be convinced) that firms that don't maximize profits will be taken over. Second, it ignores the unhappy possibility that in order to avoid being taken over, the incumbent management of a firm will pursue any number of strategies whose aim is to avoid a takeover, to the detriment of profits. At least in the United States, the financial press is filled with notions such as: "green mail," where incumbent management buys off a corporate raider (with shareholders' money, presumably) to defeat a takeover attempt; "poison-pill provisions," where incumbent management adopts policies that are solely intended to forestall raiders; and "golden parachutes," where incumbent management provides itself with employment contracts that transfer a lot of wealth to themselves and away from the firm in the event that the firm is taken over and they are discharged, thus (simultaneously) lessening the profits of the firm in the event of a takeover and lowering the incentives of incumbent management to avoid a takeover.

Formal analysis of the takeover mechanism as a form of discipline on incumbent management has been an area of very active research in microeconomic theory over the past few years. We will not review this literature here, except to give the basic observation that provided the starting point for many recent developments. This basic observation addresses the point that takeover targets will be those firms from which the raider can expect to extract profits for *herself*, which don't necessarily coincide with firms whose current management is not pursuing maximal profits.

Imagine that a firm is being poorly run to the extent that a share of its profits is worth only \$10. A raider comes along, and it is known by all and sundry that this raider will, if she succeeds in taking over the firm, adopt all manner of improvements in the running of the firm, enough so that a share in the profits of the firm will be worth \$20. To take over the firm, she must buy up a majority of the stock. But can she do so successfully? If it is known that she will succeed, then an individual holding a share of stock knows that the stock will become worth \$20. This individual

would, in consequence, refuse to tender (sell) his shares to the raider at anything less than \$20. But if this raider must pay \$20 per share, she is making nothing for herself by taking over the firm, and if there are costs to attempting a takeover (the costs of finding the target, registering the tender offer, lining up capital with which to accomplish the takeover), she will not even try. She must be compensated if she succeeds, to persuade her to try.

How might she be compensated? She might take her compensation in the form of perks: an overly generous salary for herself, a corporate jet, a corporate retreat in the Bahamas; these are hardly profit maximizing. She might benefit by diluting the value of the shares, once she takes control: If, for example, she owns a competing firm, she might be able to sell some of the assets of the firm taken over to her original firm at less than their full value. This doesn't sound like profit maximization, per se, nor is it clear that targets possessing such assets include all poorly run firms. She could benefit if she already holds a significant fraction of the shares of the firm (valued, until she shows her hand, at something less than \$20 per share); she benefits to the extent that the shares she purchased (quietly) at \$10 or so per share will become worth \$20 if she succeeds. This story hangs together rather well, as long as the raider has an opportunity to accumulate quietly a number of shares large enough so that the improvement in their value covers the costs of the takeover. But then raids that will result in minor improvements only will not pay for themselves, and while grossly mismanaged firms may be takeover targets, firms that are not so poorly managed will not be targets, and profit maximization will not be achieved by the discipline of takeovers.

We will not attempt a formal presentation of any of these ideas or those that follow them in the literature; a reference will be provided at the end of the chapter. For our purposes, the point is that justification (3) may not be entirely erroneous, but it requires a good deal of study if it is to be believed, and those analyses that have been performed do not provide unambiguous support of the justification.

- (4) (a) Profit maximization is in the best interests of shareholders of the firm.
 (b) Shareholders can and do create incentive schemes for managers that force managers to do what is best for the shareholders. So we conclude that managers maximize profits.

The second part of this argument, that shareholders are able to (and do) give managers the incentives to do *whatever* it is that shareholders want, is somewhat dubious; this should be clear from our study of

19.1. The firm as a profit-maximizing entity

incentives in part IV. At least, it should be clear that assertion (b) requires some very substantial justification itself.

Be that as it may, we take this opportunity to attack (a). Suppose that managers are just good-hearted people who will automatically do whatever is in the best interests of the firms' shareholders. Should they select whatever production plan maximizes the firms' profits?

The notion that shareholders necessarily want the managers of their firms to maximize profits is a long-standing component of the folklore of capitalism. But it is incorrect folklore, if the firm has market power¹ and if its shareholders participate, even indirectly, in the markets that are affected by the operations of the firm. Consider four shareholders of XYZ, Inc., which has market power both in output and factor markets.

Shareholder 1 is a consumer of the output of XYZ. If XYZ has and uses its market power in the output markets to push up its profits, it does so at the expense of its customers, who must pay more for what they buy. If shareholder 1 consumes a lot of the output of XYZ and owns a relatively small fraction of its shares, she can be hurt by XYZ's exercise of market power (in the pursuit of profits) more than she is helped by the distribution of those profits to shareholders.

Shareholder 2 sells factor inputs to XYZ. If XYZ exploits its market power in the factor markets to maximize profits, it may increase the dividends that shareholder 2 receives. But it may hurt shareholder 2 more by driving down the price for the factor that 2 sells.

Shareholder 3, having taken a course in portfolio management and having been convinced that diversification of her investments is the wise course of action, holds shares in many companies, some of which compete with XYZ, some of which buy from XYZ, some of which sell to XYZ. As XYZ exercises its market power, it may hurt any of those other companies (and it may help them, although hurting them is the important possibility here), and thus make 3 worse off.

Shareholder 4 buys nothing from XYZ, sells nothing to XYZ, consumes none of the factor inputs of XYZ, and holds only shares of XYZ. But this shareholder does happen to consume a good that is complementary to one of XYZ's factor inputs for some consumers (but not for XYZ itself or for shareholder number 4). Then as XYZ holds down the price of its factor inputs (in its pursuit of profits, on behalf of shareholder 4 and others), it indirectly increases demand for the factor from other sources, which increases demand for the complementary good, which bids up the

¹ Recall that if a firm can affect by its actions the prices it faces, it is said to have market power. If the firm doesn't have any effect on the prices it faces, it is said to be a price-taker or a competitive firm.

price of the complementary good, which makes poor shareholder 4 worse off.

What, then, could conceivably be the basis for this bit of capitalist folklore? Imagine that the firm is competitive. Its actions do not affect the price of any good. Now consider a shareholder in the company. Imagine that this shareholder is a consumer just as in chapter 2, who is seeking to maximize utility, subject to a budget constraint. Some of the consumer's wealth comes from the dividends the firm pays, and so the consumer is better off the more wealth she has, hence the consumer is better off the greater are the dividends paid by the firm. Moreover, higher dividends equals greater wealth is the *only* impact that the firm's decisions have on the consumer, since the firm is competitive. In particular, because the firm doesn't affect any prices, it has no impact on the profits made by other firms,^a and thus it doesn't indirectly affect the consumer's wealth adversely. Hence, in this case, the consumer is unambiguously better off when the firm maximizes profits.² What goes wrong in the four cases above is that when the firm is not competitive, the effect it has on prices can undo the beneficial effects of greater wealth for the consumer/shareholder.

We can repose this point in the context of the model of general equilibrium with firms developed in chapter 8. Recall that consumer i 's problem there was to maximize utility of consumption subject to a budget constraint which took the form

$$p \cdot x \leq p \cdot e^i + \sum_{j=1}^J s^{ij} p \cdot z^j,$$

where p gives equilibrium prices, x is the consumer's consumption bundle, e^i is i 's endowment, s^{ij} is i 's share of the profits of firm j , and z^j is the production plan of firm j . It is clear that consumer i is better off if $p \cdot z^j$ is raised (as long as $s^{ij} \geq 0$) and p doesn't change: The budget constraint loosens. But if p changes with changes in z^j , then no clear statements can be made.

We could make clear statements (or, rather, relatively clear statements) if we knew that any change in the prices faced by consumer i was "small" relative to the change in i 's income generated by increased profits for firm j . Consider, for example, a case where the only price firm j affects is the price of its own output. If consumer i is a shareholder in firm j who consumes

^a Nonprice externalities are also ruled out.

² We should have said earlier: We have in mind a world in which all profits are paid out in dividends. This isn't a realistic assumption; firms retain part of their profits in reality. This brings on multiperiod complications, for which see the discussion following.

little of firm j 's output, then perhaps we can be sure that she will prefer an increase in the profits of firm j .

One must be careful with this argument. While consumer/shareholder i may not consume much of the output of firm j , neither may she hold many shares of firm j . Very roughly put, we need to know that her "share" of the output of firm j is very much smaller than her share of j 's equity. Insofar as consumers can easily hold well-diversified portfolios (and the theory of finance teaches that well-diversified portfolios may well be optimal choices), it isn't clear how the comparison will run. For an exact analysis of all this, see Hart (1979).

We take this opportunity to add a further thought: Once we pass from a one-period, no-uncertainty world, into the world of uncertainty and many periods of economic activity, it isn't even clear what the term "profit maximization" means. The firm will hold some assets through time, and it could increase short-run profits by selling off those assets. In the interests of long-run profit maximization, we surely don't expect it to do so. The firm may be able to influence overall price levels by, for example, triggering inflation. Profits are usually measured in dollar amounts, so do we expect a firm to trigger or increase inflation, insofar as it is able, so that it will make more dollars and hence greater overall profits? How do we compare dollars earned in one state of nature against dollars earned in another? The usual pat answer to all these questions is: The firm should maximize the value of current shareholder's equity. There is a large literature concerning why this might be so (and what it means), which comes down to: This makes sense (once again) only if the firm is a price-taker, where being a price-taker in this context entails many more conditions. (For a survey of this literature, see Kreps [1987].)

All this isn't to say that firms with market power don't maximize profits, only that "because it is in the interests of shareholders" is a poor excuse indeed.

19.2. The firm as a maximizing entity

If a firm doesn't maximize profits, what might it do? Suppose we continue to take the point of view that the managers of the firm are aware of all of its production possibilities, but they do not choose a production plan to achieve maximal profits. Instead they choose in order to advance some other (personal) agenda.

What might that agenda be? To know this, we must consider, however informally, the objectives of those who control the firm. Think of managers who are distinct from shareholders. Of course, shareholders (working, one presumes, through the board of directors, which is another class of individuals whose interests and incentives must be analyzed) will attempt to provide managers with incentives to act on behalf of shareholders. So that there is no ambiguity about what is in the interest of shareholders,

imagine that the firm is competitive; shareholders would wish the managers to maximize profits. Accordingly, we can imagine that managers are given incentives to maximize profits through stock option plans, bonuses tied to the profits the firm earns, and so on.

Even so, managers are motivated by things other than their salaries.³ Managers like prestige, and it is sometimes held that the prestige a manager is accorded by her peers derives from the size of her organization measured in terms of gross sales or in terms of capital. (By what measures, for example, are the "Fortune 500" ranked?) Of course, the single-minded pursuit of ever-increasing sales or physical plant and equipment will shortly drive a firm into bankruptcy. But we might imagine a manager whose objective is to maximize some combination of profits and sales and capital; or one who seeks to maximize sales or capital subject to a minimum profit constraint.

And when there is uncertainty about profits, managers must worry about the chances of profits so low that shareholders become angry and displace them. A manager, liking her position of power and prestige, may well become conservative in the actions she takes, so as not to lose her position in some risky venture that holds out excellent prospects for profits in expectation, but that at the same time holds out some chances of losses or even worse.

Managerial theories of the firm begin with discussions of the sort above, and move from such informal discussions to models of firms in which the firm chooses a production plan, say, to maximize sales subject to a profit constraint, or to maximize capital subject to a profit constraint, or to maximize the probability that profits stay above some critical level. This sort of theory is not much in vogue these days, but it presents a first variation on the neoclassical model of the firm — firms as maximizing entities with well-defined production possibilities sets.

A variation on managerial theories of the firm (and a variation that is somewhat more in vogue at the current time) is the notion of the worker-managed firm. In this sort of model, the firm is run "in the interests of" its workers; the production plan is selected in order to maximize the size of the wage bill the firm pays, or the average compensation per worker. As with managerial theories, the firm's production possibilities are assumed known to some decision maker, who then selects whatever production plan that maximizes a well-defined objective function.

³ And it is worth observing here that it is far from clear that managerial compensation provides managers with the incentive to maximize profits. Excellent empirical work has been conducted on the structure of managerial compensation packages; see, for example, Antle and Smith (1986) and Gibbons and Murphy (1989).

19.3. The firm as a behavioral entity

To go one step further away from the neoclassical model, we can imagine treating the firm as an entity that acts in a particular fashion, but which doesn't (necessarily) act in a way that can be described as maximizing anything at all. This sort of model is especially relevant in a dynamic setting, since most interesting behavioral patterns are dynamic.

To set the scene, imagine an industry in which there are a large number of firms; six to be precise. Each of these firms hires capital and labor to produce an undifferentiated product. Demand for this product is given by the demand function $X = 100 - P$, where P is the price of the good in question and X is the industry level of supply. Because there are so many firms in the industry, firms are price-takers. In any event, the prices of capital and labor are unaffected by any decisions taken by this industry; for convenience we imagine that these prices are both one.

Production in this economy occurs in each of a sequence of periods: $t = 0, 1, \dots$. In period t , the production function of firm i depends on two parameters α_t^i and β_t^i and is given by

$$y_i = \left(\min \left(\frac{K}{\alpha_t^i}, \frac{L}{1 - \alpha_t^i} \right) \right)^{\beta_t^i}.$$

Besides the costs of the two factor inputs, each firm also has fixed costs of 100 per period.

Both α_t^i and β_t^i change with time. Specifically, suppose that $\alpha_t^i = .8\alpha_{t-1}^i + .05 + \epsilon_t^i$, where ϵ_t^i is a random variable that is uniformly distributed over the interval $(0, .1)$. And $\beta_t^i = .94\beta_{t-1}^i + .02 + \delta_t + \gamma_t^i$, where δ_t and γ_t^i are uniformly distributed over the interval $(0, .01)$. It is assumed that the ϵ_t^i , γ_t^i , and δ_t are all independent of each other. Note that α_t^i will be independent across firms, whereas the β_t^i will be correlated across different firms because of the common shock term δ_t .

(You should be able to convince yourself that if we begin with $\alpha_0^i \in (0, 1)$ and a $\beta_0^i \in (0, 1)$, then α_t^i and β_t^i for every t will live in the interval $(0, 1)$. The sequences of parameters will look something like "random walks on a rubber band"; they will drift slowly, with enough pull towards .5 so that they never drift too far away from that level. In table 19.1, you will see "typical" sequences of $\{\alpha_t^i\}$ and $\{\beta_t^i\}$ for $i = 1, \dots, 6$, beginning with $\alpha_0^i = .37, .38, \dots, .42$ and $\beta_0^i = .5$ for all i , generated by Monte Carlo simulation.)

We have in this example an industry in which the production functions of the firms are uncertain and move through time. Note that there

PERIOD 1			PERIOD 4		
FIRM	ALPHA	BETA	FIRM	ALPHA	BETA
1	.422308	.50068	1	.476185	.495287
2	.371998	.50521	2	.398806	.502979
3	.452288	.501514	3	.408959	.497389
4	.458984	.501289	4	.45912	.511639
5	.416701	.500184	5	.456818	.508136
6	.433592	.500652	6	.492404	.498119

PERIOD 2			PERIOD 5		
FIRM	ALPHA	BETA	FIRM	ALPHA	BETA
1	.466295	.495808	1	.525771	.491632
2	.41925	.501068	2	.403722	.497977
3	.439777	.496075	3	.397327	.495878
4	.478998	.504609	4	.457687	.509075
5	.422759	.499498	5	.417736	.502173
6	.491228	.49901	6	.450174	.48946

PERIOD 3			PERIOD 6		
FIRM	ALPHA	BETA	FIRM	ALPHA	BETA
1	.522857	.494829	1	.500224	.494885
2	.395073	.500286	2	.405354	.496073
3	.421801	.499734	3	.455182	.4957
4	.447678	.507847	4	.510295	.510537
5	.467395	.504796	5	.476586	.497293
6	.45029	.494539	6	.473573	.493652

Table 19.1. Sequences of $\{\alpha_i^t, \beta_i^t\}$ for $i = 1, \dots, 6$ and $t = 1, \dots, 6$.

is a "common component" to this uncertainty and an "idiosyncratic component." How would a neoclassical economist handle this sort of uncertainty? It depends on what the firms are presumed to know about their production functions. We might imagine that the firms know at each date t just what their production functions are. Then we can imagine that firms, acting as price-takers, set their output quantities to maximize profits. The supply function of any individual firm, if firms are price-takers, is given by $x^i(P; \alpha_i^t, \beta_i^t) = (P\beta_i^t)^{\beta_i^t/(1-\beta_i^t)}$.^b Note that α_i^t doesn't enter into this at all, owing to the nature of the production function and the fact that (we assume) prices of capital and labor are always equal. In a more complex

^b You might wonder how it is that firms know the equilibrium price when they set their output quantities. If this bothers you, you might think in terms of a Cournot-style model, where firms take as given the equilibrium production levels of their rivals.

example, say where the prices of capital and labor were different, α_i^t would enter into the firm's supply function.^c

In terms of our example of a six-firm industry, and the set of parameter values given in table 19.2, this would give the corresponding "equilibrium" shown in table 19.2. We'll say where these parameters come from in a bit.

One might object to this story about the firm knowing precisely the parameters of its production function. We might suppose that the firm only learns the values of α_i^t and β_i^t after it produces in period t . In this case, we could move (within the neoclassical tradition) to a model in which firms pick their production plans with a view towards, say, maximizing their expected levels of profits, given whatever information they happen to have available when production decisions must be made. This gets very complicated very quickly, and so we won't try to work out what the equilibrium would look like.

Instead, we move next to a general critique of the neoclassical approach to the problem, following the more general critique of the neoclassical approach by Nelson and Winter (1982).⁴

Nelson and Winter contend that it is unlikely the firm would even be aware of the "distribution" of its possible production function. What the

FIRM	ALPHA	BETA	CAPITAL	LABOR	OUTPUT	PROFITS
1	.53197	.49805	85.58	75.30	12.56	\$62.14
2	.50980	.50038	84.76	81.50	12.92	\$66.01
3	.51384	.49318	77.21	73.05	11.85	\$54.42
4	.55856	.49935	91.52	72.33	12.76	\$64.28
5	.50430	.49807	81.15	79.77	12.56	\$62.17
6	.52309	.49168	76.99	70.19	11.64	\$52.16

price: \$25.72

Table 19.2. Theoretical equilibrium values for given α_i^t and β_i^t .

^c We haven't said anything about what firms know about each other's production functions. In this example, what firm i knows about the values of α_j^t and β_j^t turns out to be irrelevant on two grounds: The nature of firm j 's production function is such that its supply function doesn't depend on α_j^t ; the supply of firm j does depend on β_j^t , but the price-taking assumption implies that firm i will be able to "react" to equilibrium prices, and so doesn't need to know about j 's supply function. If we had an underlying model of imperfect competition, where, say, the firms had Cournot conjectures about each other, then we would need to say what each firm knows about the other firm's β_j^t s.

⁴ What follows is a very quick summary of the arguments of Nelson and Winter. Consult their book for a fuller account.

firm "knows" (in quotes, because no single entity has this knowledge) is the pattern of productive activities that it has been able to follow in the past, its *routine*. The model given would have the firm changing the amounts of capital and labor from period to period, with changes in the two parameters. This is unlikely to happen in reality. Firms will change their production routines only infrequently. Costs are involved in shifting production routines, both in terms of hiring/discharge costs, and also (and more fundamentally) in getting members of the organization to adapt what they do. Nelson and Winter present an image of the firm as a "truce" among various employees, suppliers, and so on, wherein the conflicting interests of those parties are reconciled. While Nelson and Winter do not delve formally into the mechanics of the truce, they do note that insofar as what a firm does at any point in time is the result of such a truce, and insofar as truces, once made, are relatively hard to rearrange, firms' routines will not jump around in response to relatively minor changes in their opportunities, technology, and so on.

This is not to say that firms never change what they do. But they change relatively less frequently than the neoclassical model would predict.⁵ Firms change in response to a perception that making a change represents substantial gains. When they do change, moreover, they do not necessarily "optimize" in any exact sense. They search somewhat haphazardly for ways in which to change, and (especially) they tend to imitate the actions of those rivals that they think are doing better. (Think of all the manufacturing firms in the United States that have attempted to imitate "Japanese manufacturing techniques" over the past decade without a very clear understanding of why or how those techniques work.) Indeed, the patterns of search and/or imitation that a firm chooses to employ is itself a matter of organizational routine; if one method of search has proven fruitful in the past, then it is apt to be the method relied upon at the present, and it will continue to be relied upon until the firm feels the need to change its search/imitation routine.

In place, then, of the neoclassical pair of concepts, the *opportunity set* and *profit-maximization criterion*, Nelson and Winter offer a different pair of organizing concepts: the short-run production *routine* and the longer-run process of *search* for a better routine. They then take this pair as characterizing the single firm and embed firms in an environment where the firms will act, interact, and change according to the behavioral models posed

⁵ A true-blue believer in the neoclassical model would object that one can build into a neoclassical model the costs, if any, of adjusting the levels of factor inputs. That is, the problem is not with the neoclassical model, but with an overly simple specification of it in this context. We will deal with this objection in a bit.

and the overall environmental equilibrium. The objective is to see how the observed behavior of firms and the overall environmental equilibrium evolve together.

To take a very simple example, suppose in a six-firm industry that each firm acts as follows. Each begins with a routine that describes how much capital and labor is hired. Fixing the capital and labor quantity choices by firms, the firms' different production functions yield levels of output for each firm, and hence a market (equilibrium) price for the firms and profits for each firm. In each period, firms' profits are reported publicly, and these serve as the cue for firms to change their routines: When a firm's profits are more than 15% below the industry average, the firm searches for a new pattern of production. This search is conducted in two parts: The searching firm engages in "introspection," to see whether to adjust its capital to labor ratio. And it imitates more successful rivals in setting its level of capital input.

Specifically, a firm that is searching adopts the level of capital employed by its most profitable rival. Note that firms may be making "mistakes" by so doing; the firm being imitated may be profitable because its scale parameter β_i^j is relatively high or even because it has found a profitable (for it) capital/labor ratio. Firms recognize that they should not look to other firms for the right capital/labor ratio for themselves, so they do not engage in imitation here; instead, if a firm decides to search (i.e., if its profits are more than 15% below the industry average), it draws a noisy signal λ of the "optimal" capital proportion α_i^j . Specifically, $\lambda = \rho \alpha_i^j$, where ρ has uniform distribution from .9 to 1.1. If λ is more than 5% away from the ratio the firm is currently using, the firm adopts the ratio given by λ ; otherwise it stays with its old capital/labor ratio. (The rationale for this bit of the model is that changing ratios is costly, and firms will not undertake to make a small change.) After each period's price and profits are realized, firms decide whether to search as above. If they decide to do so, they carry out the search in time for next period's production. Parameters for production for the following period evolve according to the random process specified above, firms set their levels of inputs, and so on.

What will happen in this industry? Will firms evolve to a position where their behavior approximates the behavior of profit-maximizing, competitive firms? The system of behavior described above is fairly simple to describe, yet it seems too complex to answer even these simple questions analytically. We can, however, simulate the system. A Monte Carlo simulator for this system was built, where firms were started off "knowing" the correct capital/labor ratios and producing at the theoretically

PERIOD 98					
FIRM	ALPHA	BETA	PROFITS	SCALE	RATIO
1	.530549	.499526	173.835	110.003	.539757
2	.519665	.495387	160.707	126.692	.468657
3	.529616	.490394	156.749	118.087	.502806
4	.511884	.50331	180.407	124.876	.475472
5	.500778	.494868	180.98	132.596	.494924
6	.522573	.495059	158.708	108.484	.547317
PERIOD 99					
FIRM	ALPHA	BETA	PROFITS	SCALE	RATIO
1	.507066	.499206	172.128	110.003	.539757
2	.545456	.496198	161.179	126.692	.468657
3	.527188	.492893	170.285	118.087	.502806
4	.518491	.503433	186.755	124.876	.475472
5	.456039	.497635	182.573	132.596	.494924
6	.528068	.489151	158.599	108.484	.547317
PERIOD 100					
FIRM	ALPHA	BETA	PROFITS	SCALE	RATIO
1	.531965	.498054	172.694	110.003	.539757
2	.509804	.500375	174.758	126.692	.468657
3	.513837	.493177	168.342	118.087	.502806
4	.558562	.499345	156.798	124.876	.475472
5	.504303	.49807	187.03	132.596	.494924
6	.523086	.491682	154.07	108.484	.547317

Table 19.3. Results from a simulation.

correct scale for the starting values in table 19.1, and then 100 periods were simulated. The results of periods 98, 99, and 100 are given in table 19.3. In the first two columns we list the firms' production parameters α_i^1 and β_i^1 ; in the third column, the firms' levels of profits; in the fourth column, the total amount of capital and labor the firms hired; and in the fifth column, the proportion of total capital and labor that is capital. Note that a firm that "knows" its production function would want the first and last columns to be equal. And if the firms conform to the neoclassical model and know their production functions, they would be producing at a total scale (column 4) of approximately 156.25. More precisely, for the values of period 100, the theoretical equilibrium values are given in table 19.2.

In the three periods represented in table 19.3, no firm engaged in search, so no firm modified its production routine. Nonetheless, levels of

profits bounce around as the firms' production parameters move. In the one hundred periods simulated, fifty-one searches were conducted.

What do we learn from this simulation? This is a very simple model, built for exposition and not to conform to any realistic model of how firms operate, so the appropriate answer must be "nothing about the real world." Nonetheless, we can learn something about the model we have built. The most striking thing about the ending values are that firms are producing at a scale significantly below the scale they would use according to the neoclassical model based on the hypothesis that they know their production functions. And, in consequence, they are making higher profits than they would according to the neoclassical model.⁶ This particular simulation was run several times, and for 1,000 periods, and the scales observed in period 100 are not atypical of where firms wound up after 1,000 periods.

Can you see why this is happening? We'll give the answer in the next paragraph, but if you like to play detective, this is a good chance. The model we have described here is very seriously flawed, and the flaw, while obvious after the fact, illustrates my own lack of experience with this sort of model. Can you see where I screwed up?

Think about the evolution of the capital levels firms employ. A firm either keeps its level of capital or it copies the level employed by one of its rivals. I began the simulation with six different levels of capital, and when the first firm began to search only five were left, and so on. In the situation described in table 19.3, only two of the original six capital levels are left, and (it is relatively easy to prove rigorously) eventually only one level of capital will remain. Moreover, I began the model with six levels of capital appropriate for α s of .37, .38, .39, .4, .41, and .42, which is significantly lower than the long-run levels of α we see (which roam around in the neighborhood of .5). Hence I started the industry off in a "capital-poor" state, and I imposed dynamics for capital that ensured the industry could never escape; capital levels can never rise above the largest initial level of capital (around 65) or get any smaller than the smallest initial level of capital (around 59). Without some external source of "mutation" in capital levels that would allow a firm to experiment with a larger level of capital (and discover how profitable this can be), there is no way for the firms in this industry to attain the neoclassical scale of 156.25 once they realize that capital/labor ratios should be approximately one-to-one.

⁶ Be careful here: The profits of firm i are high because the other firms are producing at a relatively small scale. Given the scales used by the other firms and given the price-taking assumption, a firm that knows its production function would make much higher profits by increasing its scale from the values in table 19.3.

This is a very simple and flawed example of the mode of analysis adopted by Nelson and Winter. Their book should be consulted for more complete, better-analyzed, and (needless to say) much better models. But the basic constructions in this model and the use (in the end) of a simulation model to see how things evolve is typical of their general plan. It would be very useful to complement their simulations with analytical analysis, but even in the simple model given above, you can see how hard analytical analysis would be.⁷

We can tie this mode of analysis to an idea from section 19.1. Recall that one of the alleged reasons for believing that firms act as if they maximize profits is that in an evolutionary sense profit-maximizing firms will come to dominate the industry at least insofar as there is initially a "mix" of firm types, and profit-maximizing firms survive longer and grow larger. Presumably, one would build a model where firms behaved differently — some coming closer to profit maximization in their routine and in their search behavior, and some further. And one would wish to build into the model a stronger tie from period to period. For example, one could naturally model capital as a durable good, subject to depreciation, and have firms accumulate capital out of retained earnings. That way, more profitable firms would have higher earnings and, in accordance with the story we are attempting to provide, more profitable firms would grow more quickly and dominate less profitable firms. The point is that with such an elaborated model we could either give a decent burial to the story that "competition" leads to dominance by profit-maximizing firms or we could provide a setting in which the story is theoretically consistent. Some work along these lines has been done by Nelson and Winter, but a lot more remains.

This is a very new mode of analysis in economics, and it is clearly one that deserves further development in building models, running simulations, doing more exact analysis, and developing procedures for calibrating and testing models. Certainly, the sort of behavioral patterns that are naturally built into these models are more realistic than those in simple neoclassical models. A true believer in the neoclassical model would have a ready retort to all this, which would run something like: "One can accommodate within the neoclassical framework all manner of adjustment costs, search costs, and so on. The behavioral models of Nelson and Winter are entirely ad hoc. The right way to do business is to stick with the standard

⁷ Actually, the simple model we have given is amenable to straightforward analysis, but only because it is so badly flawed.

model of maximizing behavior, complicating the environment." But to accommodate all these things within a neoclassical framework is at least an awesome undertaking and probably impossible. If we are unable to analyze these models, then it is difficult at least to see how managers can truly optimize dynamically in the vastly complex problems they are set. In any event, the notion of a single manager deciding on a production plan for the entire firm does not seem very descriptive of the way firms make decisions.⁸ And we still don't have any good reason to believe in profit maximization as being descriptive of what firms pursue. Any of these would be sufficient grounds to push research in the direction of models of the sort described above. Put all together, they make the case fairly compelling: If profit maximization is a good model, then a demonstration of that fact (along these lines) should be feasible.

19.4. Firms in the category of markets

So we have firms first as profit-maximizing entities, and then as _____ (fill in the blank) maximizing entities, and then characterized behaviorally. In each case, the firm is an entity. In the justification for the sort of "inertial" behavior seen in the simulation model, in the justifications for some of the criteria that replaced profit maximization, and even in the justifications that are made for profit maximization, we see reference to the actors within the firm: managers and raiders in section 19.1, managers (and their objectives) in section 19.2, and the "truce" within a firm in section 19.3. But these are largely informal arguments. (The literature on takeovers is quite formal, and it fits into the category to which we are now coming.) When it comes down to modeling the firm formally, all these models take the firm and its behavior as the unit of analysis.

This leaves a very different way of modeling the firm to be explored, namely the firm as an arena within which the conflicting interests of managers, workers, capitalists, customers, and so on are met and resolved. But this raises the question: Why should a firm have advantages or disadvantages relative to any other "marketplace?" To address this question, and to address generally the notion of the firm (and other organizations) as marketlike entities, we turn in chapter 20 to the topic of transaction cost economics.

⁸ This isn't to say that such a model isn't a good reduced form for what does go on in a firm. But one would like either empirical or theoretical confirmation of this.

19.5. Bibliographic notes

For a survey of the theoretical work on profit maximization as an "appropriate" criterion for managers (in the sense that it is in the interests of shareholders), see Kreps (1987). For managerial theories of the firm, see Baumol (1959) and Williamson (1964). There has been quite a lot of analysis on the takeover mechanism as a restraining influence on managers; an introductory survey is provided in Schleifer and Vishny (1988). Needless to say, Nelson and Winter (1982) is highly recommended. To my own tastes, they are unduly harsh concerning the neoclassical traditions. (My reasons are given in chapter 1 and don't need reiterating here.) But the positive aspects of their research program deserve a lot of attention.

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19.6 Problems

- 1. Consider an industry in which firms produce an undifferentiated commodity for which demand is given by $X = 100 - P$. A number of

firms produce the item in question, all of whom have production function

$$Y = \left(\min \left(\frac{K}{\alpha}, \frac{L}{1-\alpha} \right) \right)^{1/2},$$

where Y is the level of output and α is some constant from $(0, 1)$, K is the amount of capital employed, and L is the amount of labor employed. The prices of capital and labor are fixed at \$1 throughout. All firms have fixed costs of \$16.

(a) Suppose precisely six firms are in this industry, all of which maximize profits taking prices as given. What is the equilibrium in this case?

(b) Suppose there is free entry into this industry, with all of the firms having the production function given above. What is the equilibrium in this case?

■ 2. (a) Suppose that firms in this industry maximize their sales revenue, taking prices as given, subject to constraint that profits must be nonnegative. What is the equilibrium in this industry if there are precisely six firms?

(b) What would be the equilibrium if three of the six firms were profit maximizers and three maximized revenue?

(c) Now suppose that all firms are revenue maximizers, taking prices as given, subject to a zero-profit constraint. Why is it difficult to contemplate an equilibrium in a case where there is free entry?

■ 3. Suppose that the industry has six firms, each of which seeks to maximize its use of capital, subject to a zero-profit constraint. (Firms continue to be price-takers.) What is the equilibrium in this case?

■ 4. To repair the model that was discussed and simulated in section 19.3, one thing that comes to mind is to have firms that are imitating the capital levels of other firms "over-do" the imitation. That is, if firm i is imitating the level of capital of firm j , firm i chooses capital that is, say, three percent higher than the level of capital of j (on the grounds that firm j is doing so well that firm i should do even more of whatever it is that firm j is doing). This is not much better (in fact, it's worse) than the model discussed in section 19.3. Why?

■ 5. Give suggestions how the model in section 19.3 might be repaired. If you have access to a computer and know how to construct Monte Carlo simulations, you might wish to try out your suggestions.

chapter twenty

Transaction cost economics and the firm

This chapter continues the discussion initiated in chapter 19 about models of the firm, with consideration given to models where the firm is something more like a market than a consumer. That is, we think of the firm as an institution within which transactions between individuals take place — an alternative to transactions that take place in a market. To make sense of this, we have to see how a transaction placed within the context of a firm is different from the “same” transaction placed within a market, which is the subject of *transaction cost economics*.

This chapter also continues to prepare for our critique of the methods that have been developed throughout the book. To accomplish this second purpose, we take things in somewhat roundabout fashion. To begin, we recount the theory of transaction cost economics as told by Williamson (1985).¹ This recounting will seem more verbal and less mathematical than topics discussed earlier in the book, and after laying out the theory along the lines of Williamson (1985), we go back and see how we can match pieces of that theory with things we did earlier.

20.1. Transaction cost economics and firms

Transaction costs

When undertaking a transaction, parties to the transaction must incur several sorts of costs. *Ex ante costs* are incurred before the transaction takes place. If the transaction is to be governed by a written contract, the contract must be drafted. Whether governed by a contract or simply by verbal commitments, the terms of the transaction must be negotiated. *Ex post costs* are incurred in consummating and safeguarding the deal that was originally struck.

¹ The rendition given here is not precisely as in Williamson (1985) in emphasis or, in some minor respects, in organization. And the rendition here is not nearly as rich in examples and elaborations as is the work being abridged.

In some cases these costs are negligible. But in other cases they can be quite substantial. Insofar as transactions can be arranged in different ways (making use of different legal and social institutions, providing more or less detail in a contract, reserving rights to the transacting parties in one fashion or another), these different ways will have distinct costs. The basic notion of transaction costs economics is that *transactions tend to be "placed" in a way that maximizes the net benefits they provide, including the costs of the transaction*. In particular, a transaction whose (transaction) costs outweigh the benefits of completion will not be undertaken at all.

To get a better fix on these costs, it is helpful to look at factors that give rise to them. Williamson compiles a list of these factors in two parts: factors pertaining to the individuals who undertake the transaction, and factors specific to the particular transaction.

Human factors

Human beings are *boundedly rational* and *opportunistic*. Bounded rationality is important because, in the first place, it means that it will be costly for individuals to contemplate and contract for every contingency that might arise over the course of the transaction; this adds to the ex ante cost of drafting the contract. These costs may be so high that the individuals fail to provide for the contingency in the contract or fail to undertake the contemplation necessary to foresee the contingency. And there may be contingencies that the individuals cannot foresee at all.² Contingencies that are not provided for ex ante may add to ex post costs, since parties to the transaction may be required to negotiate further whenever such contingencies arise. And, insofar as it is understood that some contingencies are not foreseen or provided for ex ante, the parties may build into the original contract specific means by which the contract will be amended as required. These "means" or *governance structures* bring on costs of administration and the like.

To say that individuals are *opportunistic* means that they are self-interested with guile. If it will further his own weal, an opportunistic individual will break any of the commandments. Note well the conditional *if* here. Institutions can be sometimes arranged so that when an individual gives his "word of honor," then it will not be in his interests to go back on that word to extract some short-term gain. (Think back to chapter 14 or see below if it isn't obvious to you how this can happen.)

² We will distinguish between contingencies that are unforeseeable and those that could be foreseen but that are not foreseen because of the costs of contemplation. From a theoretical point of view, these two categories could be combined by saying that the cost of foreseeing the first sort is infinite.

To distinguish simple self-interest from opportunism, think of a completely honest individual who would never break her word or misrepresent what she knows, but who still seeks to maximize her own welfare. This is self-interest, as compared to an opportunistic individual who would break his word or engage in misrepresentation under the right circumstances. Moreover, our use of the term "opportunism" is stretched to mean that it is opportunistic to refuse to divulge information that you hold and another lacks when the other person asks you to give up that information. That is, if you are selling a used car whose quality you know and a buyer asks you what the quality is, even to withhold this information would be opportunistic behavior.^a

Classify individuals according to a three-by-three scheme, where the first dimension is the individual's degree of rationality and the second is the individual's self-interest orientation.

For degree of rationality, our three categories are complete rationality, bounded rationality, and behavioral. A completely rational individual has the ability to foresee everything that might happen and to evaluate and optimally choose among available courses of action, all in the blink of an eye and at no cost. A boundedly rational individual attempts to maximize but finds it costly to do so and, unable to anticipate all contingencies and aware of this inability, provides ex ante for the (almost inevitable) time ex post when an unforeseen contingency will arise. A behavioral individual acts according to some specified behavioral pattern that doesn't (except by coincidence) correspond to the maximization of any specific utility or welfare function.

For the individual's self-interest orientation, to the opportunist and the completely honest but self-interested individuals discussed, add the utopian, someone who has a sense of the social good and seeks to maximize it.

Various pieces of economic theory can be thought of as concerning societies comprising one of the nine classes of individuals so created. If the individuals in a society are behavioral, then their self-interest orientation is irrelevant once their behavior is specified, since (by definition) they don't act out of self-interest at all.³ Also irrelevant to such individu-

^a This stretches the notion of opportunism very far indeed, far beyond its most "natural" usage. To amplify, suppose A owns a piece of land on which is buried some treasure. A doesn't know where the treasure is, and it is too expensive to dig up the entire plot to find it. B knows where the treasure is, however, and with that knowledge it would be worthwhile for B (or, if A had the knowledge, for A) to dig. According to this definition of opportunism, it would be opportunistic of B to refuse to tell A where the treasure is, if A asks.

³ While correct as stated, this is a bit misleading. Reasonable models of behavior will reflect the self-interest orientation that is supposed of the individual. For example, firms

als is all of orthodox economic theory, which is based on consumers who maximize. Evolutionary theories, such as in Nelson and Winter, but applied to the actions of consumers instead of the actions of firms, would seem to be appropriate to the economic analysis of a society of such individuals.

Most economic theory concerns completely rational individuals. If they are utopian in their interest orientation, then one employs *team theory*, in which (it is assumed) all individuals have the same utility function (which can be thought of as social welfare). They may not have common information, which makes the subject interesting, but they all act in a way that anticipates perfectly what their fellows will do contingent on the information their fellows possess, and they act to maximize some single criterion of overall welfare.⁴

Completely rational individuals who are self-interested but without guile populate the economies of general equilibrium. There is no mention of deceit or of private information. Since self-interested individuals who are not opportunistic will not withhold pertinent information, and since each side to the transaction is completely rational and so anticipates the possibly pertinent information that the other side possesses, there will be no pertinent private information at all (after a round of questions and honest answers). In undertaking a transaction, therefore, parties know just what they are getting, and any allowance for future contingencies is complete and completely understood by all.⁵

Completely rational and opportunistic individuals populate the models of moral hazard and incentives and adverse selection and market signaling; i.e., the stuff of part IV. One is able to work through how others will act and react given appropriate incentives. And one is able to work through the distribution of qualities that will be brought to market, or the distribution of qualities that accompanies some given signal or other, or the appropriate (equilibrium) action to take when faced with a menu of contracts. But the incentives that are put in place (either to take a particular action or to send a particular signal) must respect the ability of the individuals to lie, cheat, and steal if so doing would be beneficial.

in the simulation model of section 19.3 are "behavioral," but their behavior is tied to their "self-interest" as measured by relative profitability. Similarly, we could imagine a model of individual behavior where the individual responds to prompts that are tied to her "self-interest" such as her levels of consumption of various goods relative to those of her neighbors, and so on.

⁴ For an exposition of team theory, see Marshak and Radner (1972).

⁵ In general equilibrium, different firms are allowed to have different production functions. So, as a matter of the formal theory, the fact that one firm can undertake a particular production plan that another cannot is *not* based on proprietary information held by the first.

		Self-interest orientation	
		Utopian	Self-interested without guile
Behavioral	Degree of rationality	Evolutionary methods(?)	
		Team theory	Transaction cost economics
Complete	Bounded	Temporary equilibrium	Information economics
		General equilibrium	

Figure 20.1. Matching economic theories to types of individuals.

This leaves boundedly rational individuals. For boundedly rational individuals who are utopian, some pieces of team theory (Marshak and Radner, 1972) apply. For boundedly rational individuals who are self-interested but without guile, the literature of *temporary equilibrium* applies. (See Grandmont [1988].) And, finally, boundedly rational individuals who are self-interested with guile populate the world of transaction cost economics.

Figure 20.1 (loosely adapted from Williamson [1985, fig. 2-1]) presents all this schematically. (The question mark attached to evolutionary methods records the relative lack of work along these lines. The usefulness of evolutionary methods for exploring these categories seems clear, but more evidence is desirable.)

Qualities of transactions

Bounded rationality and opportunism don't come to much when the sort of transaction that is contemplated is something like an exchange of apples for oranges or for money. At least, this is so if we imagine that the buyer of the apples/oranges is sufficiently well versed in these matters to be able to tell the quality of a piece of fruit upon quick inspection. It is the conjunction of the human factors discussed above and various aspects of the specific transaction that lead to significant transaction costs. Williamson identifies three aspects or qualities of a transaction that bear on the level and nature of transaction costs: *asset specificity*, *extent of uncertainty*, and *frequency*.

A transaction has high levels of *asset specificity* if as the trade develops one side or the other or both becomes more tied to and in the "power" of the other side. A simple example is a company that makes glass bottles locating a plant adjacent to a bottler. Before the bottle maker puts his plant next to the bottler, each side can (potentially) deal with many alternate traders. If there is negotiation between bottle manufacturer and bottler

before any plants are built, we would expect (if there are many of each type) that the deal struck will reflect other opportunities that each side has. The technology of the two production processes is such that there are efficiency gains if the two locate side by side, and we expect the two to divide, according to some bargaining scheme, the efficiency gains that come from side-by-side plants. Note well that in ex ante negotiations, if a bottler doesn't like the deal that a specific bottle manufacturer proposes, she can (typically) turn to many other bottle manufacturers. But once the bottle manufacturer puts his plant next to the bottler and the bottler puts her bottling lines next to the bottle manufacturing plant, each side has specific assets at risk. Now each side has a degree of monopoly power against the other; opportunism has scope to operate. The bottler, for example, might tell the bottle manufacturer that despite the contract signed earlier she wants a lower price per bottle. The bottle manufacturer, held up in this fashion, doesn't have any as-good alternative trading partners. So, anticipating this possibility, the bottle manufacturer (and, for that matter, the bottler) may expend resources to negotiate a very rigid contract ex ante and to have the ability to enforce that contract ex post. Or, to take a slightly less opportunistic example, the bottler and bottle manufacturer may not foresee, ex ante, the impact that plastics will have on their two industries. But the two will have to adapt their transactions as the technology for plastic containers develops, and their proximity will mean that each is somewhat a captive of the other in any ex post (re)negotiations.⁵

In the story above, asset specificity is at work on both sides of the transaction. There will also be cases in which asset specificity binds only or mainly on one side. To take an example, consider the plight of the graduate student. After a year or two of study, the student has invested enormously in assets that are specific to the department at which she studies, such as meeting specific requirements, passing specific exams, and so on. Some of this may be transferable to another department, but much is not. The department, on the other hand, has much less invested in the particular student. (When the student proceeds to the dissertation stage, this balance is redressed to some extent.) What began as an exchange in which each party had many alternative trading partners becomes one in which one side has much more at risk. In such cases, relatively more resources will be expended in the form of transaction costs to safeguard the individual student from being exploited by professors and administrators; e.g., formal structures are set up by which the student can appeal unjust

⁵ For this to be an example of asset specificity, it is important that the old assets put in place earlier are not rendered completely obsolete by the new technology.

decisions to department chairpersons, deans, and the like; and both within the university and in the courts the burden of proof rests more on the university than on the student.^c

The second quality of a transaction that bears on its costs is the extent of *uncertainty* in the transaction. This goes hand in hand with bounded rationality. Indeed, uncertainty is the major complexity that gives rise to bounded rationality. Note that uncertainty here is defined very broadly. It includes uncertainty about contingencies that can be anticipated, insofar as it is costly to anticipate them or to include provisions for dealing with them within a contract. It includes uncertainty about contingencies the nature of which can only roughly (or not even roughly) be divined ex ante. And it includes uncertainty of the sort where one party has information that the other lacks.

Finally, there is the *frequency* of the transaction. This aspect of the transaction does not bear on the absolute magnitude of its costs, as with the previous two aspects, but rather on the relative costs of various means for dealing with the transaction. When a transaction between two parties recurs frequently, the two parties can construct special governance structures for the transaction, even if these special structures are costly, since the cost of the structure can be amortized over many transactions. But when the transaction is a one-time-only transaction or recurs only infrequently, then it is generally more costly to put into place specialized mechanisms for this particular transaction and relatively less costly to make use of "general purpose" governance structures, which are, perhaps, less than ideally tuned to the specific transaction.

Classifying transactions by governance provisions

Transaction cost economics lays great stress when classifying different forms of transaction on the way in which the terms of the transaction are adapted to circumstances as they arise. These features of a transaction are called the *terms of governance*. Terms of governance can be explicitly and rigidly specified within a contract that governs a transaction, for example, an explicit and formalized procedure for arbitration, as in major league baseball. Or the terms of governance can be implicit, arising from common practice and law. For example, ownership of an asset generally

^c In relations between the courts and universities, private universities are usually allowed to pursue consistent, so-called "dean's justice," where the decision of the dean is without serious appeal on procedural grounds. The notion is that if the university follows this practice consistently, then the student should know of it at the inception of the relationship, and the courts have no reason to interfere in a private transaction. Of course, this leaves the question why students would ever trust to dean's justice, a point we will return to in the next section.

confers upon the owner the right to command the use of that asset, within limits. Hence, when a bottler purchases from a bottle manufacturer the bottle manufacturing plant and simultaneously hires the manufacturer to staff the facility, the bottler has different "rights" under law than if she simply contracts to buy bottles from a manufacturer who owns the bottle manufacturing plant. The "contract" between the two explicitly specifies ownership of the assets, and the pattern of ownership together with the law implicitly specifies the resulting governance provisions.

Williamson (1985, chap. 3) gives the following classification scheme:

Transactions within the framework of *classical contracting* are those in which the terms of the transaction are completely specified ex ante. This includes the textbook exchange of apples for oranges, but goes beyond this to include any contract where adaptation beyond the explicit terms of the contract is not expected. For example, a complex purchase and sale agreement for a given piece of real estate would be included within this category.⁶ Such P&S agreements typically include very specific clauses pertaining to liquidated damages: Party 1 receives such and such if party 2 fails to perform in such and such a manner. This sort of explicit, ex ante provision for nonperformance is typical of classical contracts whenever nonperformance is an issue. Of course, enforcement of the contract remains a problem; one can rely upon bonded third parties, who act with very little discretion (such as escrow agents), and in the end (and often with this sort of contract, from the beginning) the courts are called in to adjudicate disputes.

As third parties who act with discretion are added, we move into the realm of *neoclassical contracting* or *trilateral relationships*. The contract no longer says what damages are due for various sorts of breach or, more generally, what sort of adaptation will be made in various contingencies, but instead prescribes a third party who will determine appropriate damages/adaptation, according, perhaps, to some specified procedure. For example, in arbitration one sometimes sees a scheme in which both parties to the dispute submit their "claims" between which the arbiter is compelled to choose; the arbiter may not propose a compromise.

⁶ A difficulty in providing illustrative examples in this chapter is that social customs often decide the form that a transaction takes. Real estate transactions, for example, are conducted quite differently in different states of the United States, and it is easy to envisage a society in which the form of real estate transactions is trilateral (see below), with the lending authority taking the role of the adjudicating third party. The examples that are used throughout this chapter are drawn from customs and practices that are prevalent in the United States, and they may not be descriptive of other countries. The obvious challenge in such cases is to explain in terms of the theory why there are differences. In general, cross-society comparisons of institutional practices is a very fertile field for empirical tests of the theory described here.

When the two parties to the transaction have no formal agreement about how the arrangement will be adapted to circumstances but instead rely upon their own ability to work things out as they go along, we have a *bilateral relationship*. Successful bilateral relationships remind one of cooperation in the repeated game version of the prisoners' dilemma: Each side is willing to cooperate with the other in order to preserve cooperation. (This connection will be developed more fully in the next section.)

Just as many sorts of trilateral arrangements can be made, so there are many forms of bilateral contracting. One extreme form deserves special attention. In a *hierarchical transaction* one of the two parties retains, by law or by custom, most of the authority to determine how the contract will be fulfilled. The second party will retain some explicit rights, such as to abrogate the contract, perhaps at some specified cost. And certain rights are implicitly retained under law. But up to such limits, the first party or *hierarchical superior* determines how matters will proceed. The chief example of this form, and the reason for its importance, is the classic labor contract: A worker earns his wages by carrying out the demands of his boss, retaining (in economies where slavery and indenture are illegal) the right to quit the job if these demands become too onerous or distasteful (cf. Simon, 1951).

When one party to the transaction takes command of the assets of the second, effectively internalizing the transaction, we have a *unified governance structure*. Here the focus is on the fact that "ownership" connotes control according to common practice. Note that where slavery and indenture are illegal, one party cannot buy the human capital of a second party; employment of a worker by a firm is not a transaction with a unified governance structure, but is instead a hierarchical transaction (if not some other form of transaction).

Matching transactions to governance structures

What sort of governance structure will minimize transaction costs in specific instances? Williamson advances the following scheme: Transactions are classified according to the specificity of the assets involved (from nonspecific through intermediate specificity to highly specific) and the frequency of the transaction (from occasional to frequent).

If assets are nonspecific, then there is no need for any fancy form of governance, since there is no need for complex, long-term contracts. Even if the relationship between the two parties is enduring, the relationship can be governed by a sequence of short-term contracts, since competition from the marketplace will prevent either individual from taking too great an advantage of the other. Classical contracting, then, goes with nonspecific

assets. It is when assets become more specific that the fancier forms come into play, if at all.

Two caveats to this assertion should be offered. First, this presumes that the extent to which the two parties face competition in the relationship is unchanging with time. We can imagine a situation in which A has a lot of competition initially for B, but that competition disappears through time, even though no physical specific assets are put in place. To keep the simple conclusion just given while accommodating such a case, we would need to expand the definition of specific asset to include "market power."

Second, we earlier mentioned a purchase and sale agreement for the purchase of real estate as an example of classical contracting. But when buying and selling real estate, the parties involved commit quite significant specific assets to the transaction, namely their opportunity to find in timely fashion an alternative partner should the current deal go awry. What keeps this within the domain of classical contracting is that the transaction is one with which there is a lot of experience. A real estate lawyer may be required to explain to you all the contingencies for which you must make allowance, but the extent of those contingencies is fairly well known; "boilerplate" contract forms can often be used. From the point of view of the match being developed here, what goes wrong in this case is that we do not have a third axis for "uncertainty" or "complexity." The usual real estate transaction may be very occasional (for the parties involved) and may involve substantial specific assets (for a short period of time), but the range of possible contingencies is well known; it is not complex in the relevant sense. (Or, alternatively, you can think of this as a case in which the extent of rationality is substantial, at least if the parties take the prudent course of action and employ specialists to help draft the contract.) Hence this sort of transaction can be handled by a classical contract.

Similar qualifications can be made to almost every piece of the match between transactions and governance structures that we lay out. We will refrain hereafter from making these qualifications, but be clear that this is a very simplistic scheme. It gives general tendencies and it should be qualified in many ways.

When assets are moderately specific, relational contracting (i.e., tri-lateral, bilateral, hierarchical) comes to the fore. It is the pattern of the frequency of the transaction that plays the principal role in determining which relational-contracting form is best.⁷ If the given transaction between the two parties is repeated between them frequently, then full bi-lateral contracting can work well. If one party of the two engages in this sort of transaction relatively frequently (and the second party engages in

⁷ For frequency of the transaction, you should have in mind the frequency of the specific transaction or of transactions of a roughly similar nature. Anticipating what will come later, if two parties never repeat a precise transaction but frequently repeat transactions sufficiently similar to one another so that parties can carry an overall reputation for behavior, then we would essentially have a case of frequent transactions.

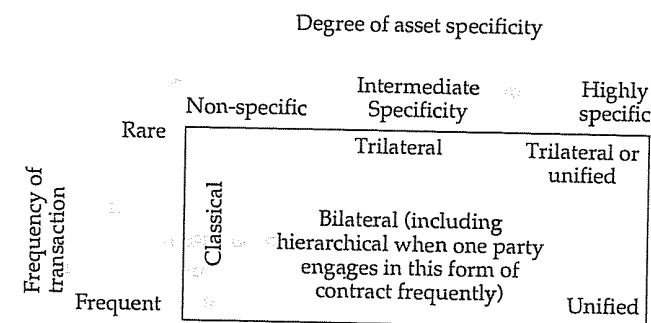


Figure 20.2. Matching governance structures to characteristics of the specific transaction.

the transaction with less frequency), then we may be able to get by with a bilateral form that gives most of the discretion to the first party; i.e., a hierarchical form may be appropriate.⁸ But as the frequency of the transaction for either of the two parties decreases, bilateral forms in general begin to suffer, and it may become necessary to enlist the aid of a third party who can act in an adjudicatory role.

When assets become very specific to the transaction, the costs of relational contracting rise: Each party has more at risk, and so must engage in more and more pretransaction planning, during-transaction monitoring, and posttransaction enforcement. A point is reached, then, where the costs of relational contracting become quite high, and the parties consider, if the laws permit, a unified governance structure, where one party buys out the other and takes full command of and responsibility for the transaction.

All these considerations lead to the picture in figure 20.2, adapted with minor modifications from Williamson (1985, fig. 3-2).

Firms and markets

The connection from the theory just developed to the theory of the firm is relatively straightforward. A firm corresponds to unified governance; a firm is a legal entity that commands an array of assets and in whose name various transactions are consummated (with other firms and with individuals).⁹

⁸ We will see one reason why in the next section.

⁹ We will not be very precise here about the differences between corporations, proprietorships, partnerships, and so on. These differences can be significant, but for a first cut at the theory they are best ignored.

Why does this make a firm into something in the category of a market? After all, the preceding sentence has firms acting in markets, playing the sort of role that is played by the individual consumer. Think of it in terms of the following example: Party A and party B wish to undertake an exchange. Party A will supply expertise concerning the design of some product, and B will supply expertise in manufacturing the product. This transaction requires certain tools — computers to aid in the design, and industrial lathes and various jigs to produce the product. We could imagine A owning the computers and B owning the lathe and jigs, in which case we would think of the exchange between them as a market-mediated exchange. Alternatively, we can imagine (say, because once B owns the jigs, he doesn't need many of the design capabilities of A) that A owns the computers, the lathes, and the jigs; she then controls a firm, which employs B for his labor services. The exchange of labor services for money is a firm-mediated transaction; as distinct from (say) the market-mediated exchange of a product design for money where B continues to own the lathes and jigs, or the market-mediated exchange where A obtains from B finished pieces produced according to plans given to B by A, which A then sells to the consumer. If A controls all the assets and employs B, it isn't important at a first level of analysis whether A owns the assets directly or controls the assets through some form of corporation; it is A's control of the assets that changes the nature of the transaction.¹⁰

In this example, if the design and/or the production technique is proprietary then perhaps the most important asset of all will be the design or the technique or both. And ownership of this asset is the most complex thing of all. If A owns all the physical capital and also the right to block others from using the design, so she controls the production amounts, and if B supplies labor services and is disbarred from taking the knowledge he acquires about the production technique to form another firm or to some competitor, then we have a very clean case of a firm (controlled by A) employing B's labor services. An equally clean case would be where B owns all the physical capital and the design, and he has the right to employ another designer who can modify A's original design, while A is barred from using the design or a modification of it. Now imagine a case where the two work across a market interface, with A owning the computer and B the machine tools. Who owns the design? Who controls the amount produced of the good? (If B purchases the design from A, then presumably B. But if B acts as a subcontractor to A, then A has this control.) Who controls the ability to modify the design? (If A sells "turn-key" jigs, then this could be A; if A sells the details of the design outright, then perhaps both have this right, depending on the details

¹⁰ The legal form of A's control may have important consequences for legal responsibilities, taxes, and the like and, at a second level of analysis, these considerations would enter into the calculus of transaction cost economics.

of the contract between them.) Even a very simple exchange like this one can become quite complex very quickly.

Inefficiencies in unified governance

This example suggests a question that brings to bear the final piece of theory that will be recounted from Williamson: What are the relative *inefficiencies* of unified governance? When telling the tale of figure 20.2, it was asserted that we move to unified governance when the level of specialized assets is high, presumably because the costs of consummating, monitoring, and enforcing the transaction at arms length, given the amount at risk and the relative infrequency of the transaction, become prohibitive. But if one avoids these costs through unified governance, then why not use unified governance exclusively?¹¹

As our example of A and B points out, the move to unified governance doesn't avoid transactions all together; instead it changes the nature of the exchange that must be made. In place of an exchange of plans for money or finished pieces for money, we have an exchange of labor services for money. This new exchange comes with an array of transaction costs itself, and it is the *relative* costliness of the various sorts of transaction that determines which form of governance we use. Note well: We see some cases of our prototypical transaction in which B buys plans from A in the market (A is an independent industrial designer), cases where A buys pieces from B (B is a subcontractor), cases where A holds all the equipment and buys labor services from B (the typical machine shop), and cases where B holds all the equipment and buys labor input from A (automobile manufacturers who employ designers). Of course, real-world examples will usually involve more than two individuals and so will be vastly more complex than the simple example we have given, but in each case you should expect to see how the relative costs and benefits of various patterns of exchange determine which pattern of exchange is used.

So what are the costs attendant to unified governance? Williamson points to the differences in what he calls *high-powered* market incentives and *low-powered* internal incentives. That is, with a full market interface between A and B, B faces strong incentives to produce efficiently, to care for his lathe, and so on. There are many things that dull these

¹¹ A related question can be asked at a more macroeconomic level. As there are such things as market power and externalities in the world, we know that a market economy will not necessarily reach a Pareto-efficient outcome. Doesn't this make a centrally planned and run economy superior? What can the market do that a centralized economy cannot? What are the inefficiencies in centralization? These questions, which have answers somewhat similar to the answers we will give for unified governance of a single transaction, are known as the Lange-Lerner controversy in the literature.

incentives when A owns all the assets and employs B. Williamson's list (1985, chap. 6), which is longer and more detailed than we can give here, but a selection follows. If A buys out B's assets and employs B, then A is unable to match such strong incentives in the employment contract she gives to B. She has a difficult time monitoring the effort B expends, and she has an especially difficult time seeing *how* B expends whatever effort he does expend. For example, A may give B a piece-rate contract, which gives B strong incentives to produce parts but very weak incentives to maintain what are now A's lathe and jigs. If B has private information about how hard a particular task is, A may have a difficult time wresting that information from him. It might be imagined that any incentives the market can provide, A can provide for B as well. But there are measurement problems in this case. If B is responsible for maintaining the lathe in good condition and if his contract gives him a financial incentive to do so, the cost of inspecting the lathe must be borne by the two parties. If B's contract incorporates some measure of the profitability of the entire enterprise, B must be concerned that A will manipulate the accounting system to B's detriment. On a more humane note, if A and B form a long-term employer-employee relationship, then A may "forgive" B if certain bad outcomes occur. But this removes the hard edge of market incentives that B would face in an arms-length, market-mediated transaction.¹²

20.2. Mathematical models of transaction cost economics

The preceding discussion, while (I hope) clear, does not conform to the style adopted throughout the book. By now you are probably used to seeing ideas explicated with models that begin with consumers who have specific utility functions, living in a particular (highly stylized) environment, who go on to maximize their way to whatever they can get. How much of Williamson's story can be cast in such terms?¹³

¹² There is another side to this last point. Sociologists are usually aghast at the way economists predicate most behavior on base self-interest, and they are quick to point out that in a "good" organization workers internalize the interests of the organization and act at least in part for it. Presumably, the social forces that lead boss A to forgive and protect subordinate B also lead subordinate B to act against his own narrow interests and in favor of the interests of A. Forgiveness may be a cost of unified governance (or any form of long-term relationship), but it may accompany substantial benefits.

¹³ Why should we attempt to recast Williamson's story in this fashion? The maintained hypothesis throughout this book is that formal mathematical modeling promotes comprehension and clarity of thought, and it is especially valuable for checking the internal consistency of the stories told.

Williamson locates transaction cost economics in the domain of individuals who are boundedly rational and opportunistic. After part IV (and even part III), you should have no problem with opportunism. But all the consumers we have analyzed in this book have been "rational," at least in the sense that their choice behavior has conformed to the basic models of chapters 2, 3, and (when there were dynamics) 4. How are we to model boundedly rational behavior? What does boundedly rational behavior mean?

As noted in part I, there are no generally accepted answers to these questions. There has been some movement towards developing theories of boundedly rational behavior, but nothing yet has taken center stage as the standard model. In retelling pieces of the preceding story in terms familiar from earlier parts of this book, we must rely on models of individual behavior that are rational in all the standard ways. In effect, we are pushing the theory of transaction cost economics into the domain of individuals who are unboundedly rational and opportunistic. It is important to think through how this colors the analysis and, as importantly or more so, what is missed by this.

With all this as prologue, we mention (in varying degrees of detail) a few of the pieces of the grand scheme of the previous section that have been scrutinized in more mathematical terms.

Incomplete contracts and ownership

Grossman and Hart (1986) analyze a variation on the basic story of the designer A and craftsman B. Their focus is on who should own which assets. The story runs roughly as follows: A and B must, in a two-period model, decide how much to invest in their particular assets, A in a computer system that will aid the design and B in a lathe and jigs. We let a_A and a_B be these levels of investment. The investment levels are chosen, and then further decisions q_A and q_B must be taken. The benefit to party X ($X = A$ or B) depends on the initial decision a_X that X takes as well as the two subsequent decisions and is given by a function $V_X(a_X, q_A, q_B)$.^d

The crucial assumptions made in the analysis are:

- (1) At the outset, before the decisions a_X are taken, the two parties can negotiate over ownership of the two types of assets. This is done in a competitive market. Ownership of an asset gives its owner the right to choose q_X . So, for example, if B sells his assets to A, then in the second period A chooses both q_A and q_B .

^d Grossman and Hart use a particular functional form for V_X . As we won't attempt to reproduce their analysis but only to relate the form of their story, we won't go into these details.

- (2) Irrespective of who owns the assets, party X must choose a_X . Think of a_X as a maintenance decision, made in an earlier period while the assets were being used.
- (3) Payments made for the exchange of ownership rights to the assets cannot be made contingent on the a_X , q_X , or the values of V_X .
- (4) Binding contracts concerning the levels of the a_X cannot be written.
- (5) Binding contracts concerning the levels of the q_X can be written, but only after the a_X decisions have been made. It is assumed that efficient levels of the q_X will be chosen. If each party retains ownership of his own asset, then the gains from choosing efficient levels of the q_X (over the levels of benefits obtained if the q_X are chosen noncooperatively) are split 50-50 between the two, coming from some (unmodeled) bargaining process.

Assumptions 3, 4, and 5 drive the model and deserve comment. Assumption 4 corresponds to a standard story of moral hazard with no observability at all. This also explains why it is that the prices charged for ownership rights cannot depend on the a_X . It is further assumed that the V_X are not observable or, if observable, cannot be made the basis of an enforceable contract. (The latter possibility can arise if the levels of the V_X cannot be verified to a standard required by a court of law or some other third party who would be called in to adjudicate any disputes.) As for the q_X , the notion is that q_X comprises a lot of detail about what to do with asset X , depending on contingencies that ex ante cannot be foreseen, or that are too expensive to foresee, or that are foreseen but too expensive or (because of problems of verifiability) impossible to provide for contractually. Ex post binding contracts concerning the q_X can be made, however, since once the contingencies have been realized, a simple, single decision is taken.

When Grossman and Hart turned to analysis, they assumed that the two individuals are well aware ex ante what levels of q_X will be chosen (depending on the pattern of ownership rights). That is, there is no uncertainty about the level of q_X in the model. Why is it infeasible to contract over the levels of the q_X in the first period, but feasible to specify these levels contractually in the second period? Grossman and Hart suggested a model in which q_X is an overall level of investment or effort, which must be directed in particular ways ex post, and one cannot specify ex ante which of the many ways investment or effort should be allocated because of an inability to foresee later contingencies. But then why not write a contract giving the overall level of investment or effort in the first period?

A consistent story is that while an enforceable contract *can* be written concerning the precise allocation of investment or effort (so much to task 1, so much to 2, and so on), one cannot write an enforceable contract on the overall level (so much effort to be distributed in a manner later to be determined). And, in the first period, a detailed contingent contract is either impossible or too expensive to write.

Because ownership confers the right to choose the q_X , different patterns of ownership yield different initial decisions concerning the a_X . If A sells her assets to B, she can anticipate that q_A and q_B will be chosen with B's interests in mind, and this will affect her choice of a_A . A similar story results if A owns the assets. And if each party owns his own asset, then the a_X decisions are made with a view towards strengthening the bargaining position of each, in the bargaining envisioned in (5). Grossman and Hart show how different parameterizations of the model lead to different patterns of ownership being optimal; in effect, they give cases where it is optimal for A to buy out B, for B to buy out A, and for the two to maintain an arms-length, market-based relationship.

This model gives a mathematical look into several of the aspects of transaction cost economics crucial to theories of the firm. The opportunism of individuals is represented by the choices of the a_X . Asset specificity is captured by the fact that each side is locked into a relationship with the other once we move past the stage of choosing the a_X . Governance ex post is determined by ownership; the contract is "filled out" in the second period according to who owns the assets. If we make a rough identity between ownership of assets, unified governance, and firms, then we have a theory that, depending on the particular circumstances, specifies what is the transactionally efficient scope of the firm or firms.

The importance of bargaining costs

Milgrom and Roberts (forthcoming) offer an important commentary on the basic scheme of transaction cost economics. Specifically, they reinforce the importance of the human factors to the story, and they suggest that emphasis in the context of market-mediated transactions should be placed on *short-run bargaining costs*. Their argument runs as follows: Imagine two parties to a transaction are fully rational (in particular, they can foresee all future contingencies and how those contingencies will be met, even if a contract doesn't specify these things) and are able to execute binding short-term contracts. Assume as well that the parties are risk neutral, so that any redistribution of wealth between them has only distributional consequences; there are no income effects.

In general, if the two parties undertake a long-term transaction and do not write (for whatever reason) a long-term contract covering all contingencies, they will have to engage in bargaining and renegotiation as events unfold. In general, bargaining can be costly; time is taken, and (although we didn't see this in the simple settings of chapter 15) in cases where parties have private information, inefficient outcomes can result. It is clear, therefore, that bargaining costs are an important part of transaction costs.

Milgrom and Roberts argue that in this context bargaining costs are the *only* part of transaction costs. Even if there is high asset specificity, much uncertainty, and any specification of frequency, if short-term bargaining costs are zero, all transaction costs will be zero; the parties will reach efficient arrangements as time unfolds. What does it mean for short-term bargaining costs to be zero? That binding contracts governing short-run actions can be made at zero cost, and these contracts always result in short-run efficient actions (holding fixed subsequent actions).

Consult Milgrom and Roberts (forthcoming) for the formal argument, but a sketch is easy to give. In the final stage of any transaction, the assumptions guarantee that an efficient arrangement will be reached. Move back to the penultimate stage. Because the two parties are farsighted, they know what arrangements will arise in the final period, and they know those arrangements will be efficient. Because they are risk neutral, any redistribution of wealth that will take place in the final round of arrangements can be "undone" at the current stage.^e Applying the no-bargaining-costs assumption again, they achieve an efficient arrangement concerning the penultimate round of actions, and so on. They can work back to the start of the transaction, and their short-term agreements will guarantee efficient actions all the way along.

All of Milgrom and Roberts' assumptions play a role in this argument, but two deserve special highlighting. The assumption that the two parties are rational (in particular, farsighted) is crucial, and it reinforces the importance Williamson attaches to human factors. The assumption that enforceable short-term contracts can be reached is crucial. Compare the conclusion reached by Milgrom and Roberts with that reached by Grossman and Hart. Grossman and Hart had rational, farsighted individuals. But in their story, these individuals cannot sign a binding agreement concerning the initial round of decisions a_X . As both Grossman and Hart and Milgrom and Roberts noted, if the a_X could be made the subject of a

^e If they discount, they would seemingly have to discount at the same rates or have available simple loan contracts between themselves so that they could take advantage of any efficient trading of wealth across time.

binding contract, then (in their setting) the ownership of the assets would be irrelevant.

Having put bargaining costs into the spotlight, Milgrom and Roberts went on to study the costs of bargaining and their sources. In so doing, they made use of the formal literature on bargaining — what we discussed in chapter 15, what we omitted (i.e., bargaining with incomplete information) — and also work on "bargaining via particular mechanisms," such as auctions. And they discussed how both asset specificity and measurement problems play a role in bargaining costs.¹⁴

Chapter 14 and figure 20.2

A third way we can set the discussion of the first section to mathematics is to interpret figure 20.2 using the theory of repeated play from chapter 14. We saw in chapter 14 that cooperative behavior can arise in situations of repeated interaction where each side to the transaction cooperates because each has a stake in the future sufficient to outweigh any gains that could be obtained by acting opportunistically in the short run. The theory of chapter 14 suggests the following:

- (a) Each side must have a stake in a maintained relationship. Or, more accurately, if one party is at risk from opportunistic behavior by the second, then the second must be at risk in the future if he engages in opportunism. Moreover, if the short-run gains for the second party from opportunism are quite large, the current value of the long-run gains from a continued relationship must be large as well. Either the relationship must be repeated fairly frequently (so that, effectively, the discount factor is close to one), or the value of the continuing relationship must be high.
- (b) It is enough that the party who puts the other party at risk repeats the transaction fairly frequently, has a substantial amount at stake in the execution of any single transaction, and has her performance observed by potential future trading partners. Then this party's behavior can be tied to her reputation for behaving in a particular way.
- (c) The noisier the environment, the less well such constructions work.
- (d) There are many possible "equilibria" in repeated interactions, so that definite predictions can become difficult.

Figure 20.2, at least in one respect, appears to fit well with this theory. As the frequency of the transaction is lowered, the nature of relational

¹⁴ They went on to study the sources of "bargaining costs" in unified governance, to which we will come in a bit.

contracting passes from bilateral and hierarchical to trilateral. This can be explained in some cases very naturally and directly from (a) and (b) above.¹⁵ Think, for example, in terms of the sale of securities. An individual seller of securities may sell with sufficient frequency to particular buyers (or a particular community of buyers) so that no intermediation is necessary. An example might be AT&T directly marketing its securities to its shareholders (although even here there is some intermediation). But in most cases a third party, an *exchange*, is brought into the picture; the exchange provides certain guarantees that facilitate trade. For example, in futures contracts, the exchange acts as guarantor that the contract will be made good. The exchange polices the trades that take place, monitoring the actions of professional traders, and the exchange (often) offers arbitration services when disputes arise. For this the exchange is paid some sum of money. It is the stream of those payments (or, rather, the amount by which this stream exceeds the direct costs of the service) that gives the exchange an incentive to police trades made in its name, at least as long as investors monitor the reputation of the exchange and are willing to take their business elsewhere if the exchange doesn't fulfill its part of the bargain.¹⁶

Another example of the applicability of (a) and (b) to the study of governance of transactions is given by Simon (1951). He considers the basic labor exchange relationship and, in particular, the fact that the employer is typically a hierarchical superior in this relationship. That is, the employer specifies what tasks the worker will undertake as those tasks arise, subject only to some general "rules" and to the worker's right to quit. Discretion is reserved largely to the employer, and so the reputational glue that holds this together (that allows the worker to trust the employer) is the general reputation of the employer.

¹⁵ Other explanations can be given as well. For example, the lower the frequency, the more the cost of a specialized institution must be amortized over a single transaction, and hence the greater the incentive to move to some form of "general purpose" trilateral relationship. See Milgrom and Roberts (forthcoming) for a detailed development of this argument.

¹⁶ Reputation is not the full story here. (1) Informational efficiency plays a role. Because many trades take place on the exchange, the exchange is able to centralize information about the traders — who is honest and who is not. Even if individual investors can obtain information about one another and deal with each other with sufficient frequency that bilateral relationships are possible, the relatively lower costs of the information for the exchange may still push the transaction from bilateral to trilateral form. (2) Securities trading is really a good deal more complex than a simple exchange between two individuals. Typically, each individual has an agent, a broker, who executes the trade for the individual. As individuals deal relatively infrequently with their brokers, and brokers deal with each other constantly, the possibility of collusion between the brokers against the interests of the individuals is quite high. Hence the exchange has a role to play in monitoring the actions of brokers.

Point (c) fits very neatly into the commentary that accompanies figure 20.2 as well. Quoting from Williamson, successful bilateral relationships are found in cases where the conditions to which parties adapt are "exogenous, germane, and easily verifiable ... and ... consequences must be confidently related thereto" (Williamson, 1985, 77). In other words, it must be clear to each party that circumstances call for an adaptation, and the nature of the required adaptation must also be clear.

Consider in this context the trilateral relationship between a publicly held firm, investors, and external auditors. The firm, to raise capital provides investors with information about its financial health. Since a good deal of moral hazard is associated with the provision of this information, a third party, an auditor, is brought in to bless the reported information (or not). These auditing firms are well paid for their services, as long as they maintain a reputation for giving honest audits. The question is, On what is this reputation pegged? Because each audit engagement is unique and because the number of decisions an audit firm must make is very large, if each audit was tailored to very specific circumstances, it would be quite difficult to decide *ex post* whether an audit firm was diligent and honest in what it did. Accordingly, when the quality of an audit is questioned, what is important is whether the auditing firm followed well-established guidelines (in the United States called Generally Accepted Auditing Standards) that are set by the industry, even if following those guidelines does not provide the most informative audit possible in the particular circumstance. Moreover, auditors provide very coarse signals of the information they receive. In most cases, they say only that the information provided by the firm is "basically okay" by giving an "unqualified" positive opinion. By having fixed procedures and very coarse reports, it is possible to check *ex post* whether the auditing firm did what it is supposed to, and so it is possible to maintain a useful reputation as an independent third party.

It is worth noting that chapter 14 doesn't apply precisely to this story. Chapter 14 concerns simple repeated game situations where one party (or both) plays the same game over and over. In most applications, a transaction is never repeated precisely. But you shouldn't have a hard time seeing how the theory in chapter 14 adapts to cases in which the sequence of "repeated transactions" is really a sequence of similar but not precisely identical transactions. As long as one can find some rule that can be applied (in observable fashion) to the sequence of similar transactions, reputation and folk-theorem style constructions can be made to work.

While the theory of chapter 14 helps us to understand parts of figure 20.2, it does not explain all of that figure. In particular, for very high levels of asset specificity, figure 20.2 calls for unified governance. How do we

explain this? (1) Perhaps it has something to do with the requirement that the value of a continued relationship must rise with increases in the short-run inducements to act opportunistically. With higher levels of asset specificity, short-run inducements can certainly be higher. (2) As assets become more specific, losses from the loss of cooperation increase. Insofar as there are noisy observables, so that relational contracting will require some periods of "noncooperation," we might expect to find a move towards unified governance as the cost of those periods increases.^f (3) There is also intuitive appeal to the notion that "strategic risk aversion" plays a role here. Parties to a relational contract may be afraid of finding an inappropriate trading partner, of a relation gone permanently sour, or of long periods of time spent sorting out how the benefits of cooperation will be divided; higher levels of asset specificity mean an increase in the cost of such things, and thus an increase of the overall costs of a relational transaction. Some of these things can be found in chapter 14, but others play no role there; it remains to refine the theory of repeated interaction to encompass the missing effects.

Firms and reputation

Firms have no role to play in the previous subsection, but it is not difficult to provide one for them. In all the constructions of chapter 14, at least one party must be long lived. There are long-lived individuals for whom these constructions can work, but the reach of these constructions is lengthened considerably if we allow firms to carry reputations. A stylized model of the basic idea runs as follows. We imagine a sequence of pairs of individuals who play the game in figure 20.3. That is, at date t , for $t = 1, 2, \dots$, an individual A_t plays this game against B_t . Note that the unique Nash equilibrium has B choosing x' , which gives both sides a payoff of 0.

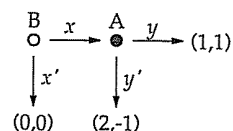


Figure 20.3. An extensive form game.

In this game, the payoff to player A is listed first and that of B is listed second.

^f Note that these factors would bind relatively less strongly on the third party to a tri-lateral arrangement, and so we do not completely lose this form of governance for highly specific assets and low frequency.

If we had one player A playing a sequence of B s, and if this A discounted her payoffs with a discount factor close to one, in the spirit of chapter 14 we could construct an equilibrium in which B_t always chooses action x and A chooses action y , giving both sides a payoff of 1: A chooses y because if ever she chooses y' , all subsequent B s choose x' . In effect, A maintains a reputation for choosing y .

But if each A_t plays only once, this construction will not work, and the mutually beneficial transaction x followed by y cannot be supported (owing to A_t 's opportunism and inability to guarantee that she won't select y').

We can get back the reputation construction if we imagine that each A_t lives and consumes for two periods. In period t , A_t plays the game shown in figure 20.3 against B_t . In period $t+1$, A_t retires and lives off her savings. Assume that A_t 's utility is given by $c_t + .95c_{t+1}$, where c_k is A_t 's level of consumption of the numeraire good in period k ($k = t, t+1$).

Suppose then that a "firm" is formed by A_1 , called *Honest A, Inc.* This firm carries a reputation with the B_t s as follows: B_t is willing to trust A_t (that is, play x) if A_t owns *Honest A*, and no previous A_s who owned *Honest A* ever chose y' when trusted by B_s . That is, if some A_s ever chooses y' instead of y given the chance, and if this A_s owns *Honest A*, then the reputation of *Honest A* is irretrievably besmirched.

If the B_t s act in this fashion, each A_t has a positive incentive to purchase *Honest A* from A_{t-1} if the reputation of *Honest A* is still good. Suppose the purchase price of *Honest A* is 10 as long as its reputation is good. Will A_t purchase this firm from A_{t-1} ? If A_t believes she can sell the firm to A_{t+1} for \$10 as long as she doesn't hurt its reputation, the answer is yes. Her choices are:

- (1) Don't purchase the firm from A_{t-1} . Then B_t will choose x' , and A_t will obtain 0 in each of periods t and $t+1$, for a discounted payoff of 0.¹⁷
- (2) Purchase the firm, and then, when B_t chooses x , choose y' . This gives $2 - 10$ in period t (the payoff from the game less the purchase price) and 0 in period $t+1$ (her ownership rights are worthless, since she sullied the reputation of the firm), for a discounted payoff of -8 .
- (3) Purchase the firm, and then, when B_t chooses x , choose y . This gives $1 - 10$ in period t and then 10 in period $t+1$, the latter being the proceeds from selling the firm to A_{t+1} . This gives a discounted payoff of $.5$.

¹⁷ We do not include in these computations the utility A_t derives from any endowment she might have. Since we have assumed a linear utility function, this is okay. We should assume that A_t has endowment of at least 10 in period t , so she can afford to buy *Honest A* from A_{t-1} .

an inability to commit a priori to a particular incentive scheme and/or intervention rule.

We have seen this theme (the inability to commit) in several places in the book, and we will not go into much detail here. But, to refresh your memory, recall our discussion of the ratchet effect. We imagine a principal, an agent, and a task the difficulty of which is unknown to the principal. This task must be performed twice (say), and (we suppose) the principal cannot commit to a two-period incentive scheme. If in the first period the principal learns from the agent's performance precisely how difficult the task is, then she will use this information in designing second period incentives. Hence it will be in the interests of the agent to make the task look harder than it really is in the first period. Note that the root of this problem is the principal's inability to commit to a two-period incentive contract. If the principal could commit at the outset to a compensation package (and to refrain from manipulating the accounts, intervening whenever she wishes to ex post, forgiving unlucky agents, and so on), then she can (theoretically) match high-powered market incentives.

We can take these ideas a step further with an analysis of influence costs. In a unified governance structure, the authority to make decisions is centralized, and individuals affected by these decisions will wish to influence the central authority to whatever extent is possible. If the central authority can be corrupted, attempts will be made to corrupt her. If activities intended to corrupt the central authority take time away from more productive activities, this is a cost of unified governance. These costs increase if the corruption causes the central authority to make decisions that are inefficient. Of course, corruption across a market interface is not unheard of, but the more centralized the authority, the greater will be the level of energy devoted to corrupting activities.

It might seem that this depends on a corruptible centralized authority. But even an "uncorruptable" central authority may be amenable to influence. Insofar as the central authority lacks information relevant for making decisions and looks to subordinates (or to their performances) for this information, then those subordinates will attempt to influence the central authority by manipulating this information. The central authority may well be aware of these attempts at manipulation, but the only way to stop them entirely is to shut off the flow of information, which may be far from optimal. In equilibrium, even with an uncorruptable central authority, wasteful influence activities may go on. See Holmstrom (1982), Holmstrom and Ricart i Costa (1986), Milgrom and Roberts (1988), and Tirole (1986) for formal models.

20.3. Bibliographic notes

Repeated reference has been made to Williamson (1985) as the basic source of transaction cost economics; this is an important summary statement of the subject that should be read by all. While Williamson's contributions have been many and important (not the least of which is the unified treatment of the subject in his book), the subject really originates with Coase (1937). Chandler (1977) and Klein, Crawford, and Alchian (1978) provide other important perspectives.

As for the more mathematical analyses of section 20.2, besides the already referenced Grossman and Hart (1986), Holmstrom (1982), Holmstrom and Ricart i Costa (1986), Milgrom and Roberts (1988, forthcoming), Simon (1951), and Tirole (1986), see Kreps (forthcoming) for the notion of a firm as the carrier of a reputation that passes from one generation of owners to the next and Wilson (1983) on the role and nature of reputation in auditing. For a survey of other work along these lines, see Holmstrom and Tirole (1989).

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postscript

As we compare transaction cost economics in section 20.1 with the more mathematical attempts at theorizing related in section 20.2, several pieces missing from the mathematical analysis spring to mind:

- (1) Williamson (1985) stresses bounded rationality and, especially, the inability of individuals to foresee all future contingencies and how they will be met. The argument of Milgrom and Roberts (forthcoming) shows just how important this can be.
- (2) We rely on folk-theorem constructions to motivate relational contracting. But as we saw in chapter 14 such constructions alone do not give particularly sharp predictions. There are many plausible equilibria in repeated games situations, and the theory doesn't give us much of a handle on which will be adopted. And if we cannot pin down which equilibrium will be adopted (or, rather, if we cannot think why some specific equilibrium will be adopted by the participants), then there is no particular reason to restrict attention to equilibrium outcomes. We certainly have a problem if we try to analyze how the equilibrium will shift with small changes in the environment. (In fact, Nelson and Winter [1982] have suggested that small changes in the environment will be met with no change at all in the equilibrium. Similar predictions are made in the literature of sociology and social psychology. See, for example, Hannan and Freeman [1984] and Stinchcombe [1965])
- (3) A similar problem confronts us if we think of parties bargaining (in the short run) in a situation where each has a degree of monopoly power over the other. We saw in chapter 15 that one can get quite specific predictions from bargaining models with certain types of bargaining institutions, but the predictions seemed intuitively to be too sharp, intuition that is verified in experiments. Individuals may bargain to consistent and definite outcomes in certain situations, but the extant formal theory doesn't seem to be of much use in predicting what those outcomes will be or how they will change with small changes in the environment.
- (4) Earlier in the book, we complained that a weakness of game theory was that it took institutions (the rules of the game) as given exogenously, without explaining where the institutions come from. We can view transaction cost economics as giving us a bit of a lead on this, insofar as we

maintain the hypothesis that institutions will be created with a general view toward minimizing transaction costs (or, more precisely, with a view towards maximizing the net benefits of transactions). But important pieces of transaction cost economics depend on the society in which the transaction is embedded — on the framework of laws and customs, in particular.

In various places in this book, I have advanced the notion that the solution to the problem of multiple equilibria, pointed to in (3) and (4), lies in the individual's strategic expectations and social and normative environment. This thought is certainly not original to me; the heart of Schelling (1980) is predicated on this; it is evident from much of the experimental work in bargaining, such as Roth and Schoumaker (1983)¹; and it can be found in the literature of sociology (Granovetter, 1985; Parsons, 1954). But when one approaches the problem of strategic expectations with the choice-theoretic orientation of traditional microeconomics, a barrier is hit: Many things *can* be an equilibrium, if we arrange everyone's a priori expectations correctly. There are no limits to the individual's processing power. There are no theoretical reasons to expect players to pick a simple equilibrium. (We don't even have a generally accepted formal definition of the term "simple.") For example, we can imagine repeated game equilibria in which play at date n depends on the n^{th} digit of an octal expansion of π ; players in our models can anticipate that this is how their fellow players will behave, and then they respond in kind.

To my mind, all the items listed above are tied together, and their resolution will come only with better models of boundedly rational choice; in particular, boundedly rational choice in dynamic settings, where individuals do not have the incredible powers of foresight that they are granted in the standard models. Instead, individuals learn (or think they do) somewhat imperfectly from past experience about what will happen in the future. It is this retrospection that forms strategic expectations, and, moreover, it is experiences from the past that (imperfectly) guide the evolution of institutions.

The game of chess provides an interesting example. Played by the sort of individuals who populate the economies of this book, a chess match would be a boring affair; the players would come on stage, look at each other, and agree to whatever (forced) outcome the game happens to have. Of course, because of the complexity of the game, no one knows what outcome is forced. And so players are guided by experience and (very occasionally) by experimentation: This or that variation or gambit seems

¹ See also Roth (1985).

to work in this or that way, where the evidence is that this is how things have worked in the past. But no player can be completely sure how a particular move will work, and it is especially exciting when grandmasters suddenly return in a match to "well-known" variations and try to find advantage where none is thought to exist.

From the point of view of this book, chess is a dangerous example to mention. Only the simplest of endgame situations fit nicely into the sort of equilibrium analysis we have used here. If the economic environment is as continuously complex as chess, then we would seem to have wasted a lot of time studying equilibrium techniques. But (of course) the implicit assertion is that economic situations are not in all respects so complex that equilibrium analysis is irrelevant. Pieces of the economic environment, parts of the greater "game" each economic actor plays, are amenable to the form of analysis we have used. The problem comes in applying the forms of analysis we have used for larger chunks of the game each actor faces; especially chunks that involve long stretches of time and much uncertainty. Moreover, we may have a good intuitive feel for when equilibrium analysis is useful (on small pieces), but our feel depends on there being (roughly) common strategic expectations about how to behave in a particular situation, and we have much less handle on how to model where those strategic expectations come from or how they change. The techniques we have are useful in some contexts, but they are potentially dangerous in others, and even when they are useful, we don't have the complete story why. It remains to develop the tools necessary for the bigger picture and to place the smaller picture properly in that bigger picture.

These observations are also hardly original to me. A long list of economists and social scientists have expressed this view both earlier and with greater eloquence than I. The difficulty, also generally recognized, is in finding acceptable tools, which I take to be models of boundedly rational behavior (and especially models that stress retrospection).

There has recently been an upsurge in interest in formal models of bounded rationality, but since no one of the models advanced is yet emerging as a "winner," I will not go into details.² So I close simply, with the cautions expressed above and with the hope that in subsequent editions of this book (if any appear), this postscript will be gone, and the material on dynamic choice, game theory, and firms and other institutions will look quite different and will be more satisfactory.

² Since I have my own entries in the sweepstakes, I am probably too biased to give a good survey anyway.

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appendix one

Constrained optimization

The aim of this appendix is to give you a bare working knowledge of how to solve problems of the form

$$\text{maximize } f(x)$$

$$\text{subject to } g_i(x) \leq c_i \text{ for } i = 1, \dots, n$$

for functions $f, g_1, g_2, \dots, g_n$ whose domain is R^k for some integer k and whose range is R , and for constants $c_1, c_2, \dots, c_n$. The mathematical theory of optimization with inequality constraints concerns all the things discussed here, and this discussion won't get you within shouting distance of that theory. So while this appendix will tell you enough to make it through the applications in the book, you should engage in further rigorous study of this material; two recommendations are made at the end of the appendix.

We'll proceed as follows. First, in cookbook fashion, the recipe for finding solutions to this sort of problem will be given. Then an example will be worked to illustrate the recipe. And then, by looking at the simplest possible example we'll try to give some intuition for the recipe.

A1.1. A recipe for solving problems

We want to solve

$$\text{maximize } f(x)$$

$$\text{subject to } g_i(x) \leq c_i \text{ for } i = 1, \dots, n.$$

Here, mechanically, is what to do.

Step 1. Form the Lagrangian.

For each of the n constraints, create a *multiplier* — this is a real variable. The multiplier for the constraint $g_i(x) \leq c_i$ will be denoted by λ_i . Then the Lagrangian is the function

$$f(x) - \sum_{i=1}^n \lambda_i g_i(x).$$

(In some books the term for the i th constraint is $\lambda_i(g_i(x) - c_i)$. It will make no difference for the form of the recipe that I'm giving you here.) This should be thought of as a function of $k + n$ variables, namely the k components of x and the n λ s. We'll write $L(x, \lambda)$ for this function, where λ should be thought of as the vector $(\lambda_1, \dots, \lambda_n)$.

Step 2. Write out the first-order conditions for the x_j s.

The first-order condition for the variable x_j ($j = 1, \dots, k$) is that $\partial L / \partial x_j = 0$, or, expanded,

$$\frac{\partial f}{\partial x_j} - \sum_{i=1}^n \lambda_i \frac{\partial g_i}{\partial x_j} = 0.$$

Step 3. Write the n constraints.

Not much to do here: The i th constraint is

$$g_i(x) \leq c_i.$$

Step 4. Write the inequality constraints for the multipliers.

This is purely according to recipe; the multipliers must all be nonnegative, or

$$\lambda_i \geq 0.$$

Step 5. Write the complementary slackness conditions.

There are n of these — one for each constraint. The i th complementary slackness condition is that

$$\lambda_i(c_i - g_i(x)) = 0.$$

Stare at this one for a moment. In step 3 we required that $g_i(x) \leq c_i$, or $c_i - g_i(x) \geq 0$. In step 4 we required that $\lambda_i \geq 0$. Hence the product $\lambda_i(c_i - g_i(x))$ is nonnegative. Now, in this step, we require in addition that the product be zero. That is, we require that if the i th constraint does not bind (that is, if $c_i > g_i(x)$), the i th multiplier must be zero. And if the i th multiplier is positive, the i th constraint must be an equality (that is, the i th constraint must bind). Note that we don't preclude the possibility that the multiplier is zero and also that the i th constraint binds; both terms in the product could be zero.

Step 6. Mix all the above ingredients.

Look for a solution in x and λ to the first-order conditions, the two types of inequality constraints, and the complementary slackness conditions. When and if you find one, it is the solution to your problem.

That's it. That is the recipe for finding the solution to constrained optimization problems. For reasons alluded to below (that are properly explained in good books on optimization and nonlinear programming), this recipe will produce a solution to most of the constrained optimization problems in this book. In following this recipe, it is crucial that you get the "signs" right; this recipe works: (a) if the problem is a maximization problem; (b) if the constraints are stated in the form of $g_i(x) \leq c_i$; (c) you are careful to write the Lagrangian precisely as above (note the minus sign in front of the sum term); and (d) you remember that the multipliers must be nonnegative.

How do you make the recipe work in variations? If you are asked to minimize $f(x)$, then that is the same as maximizing $-f(x)$. If you are asked to satisfy a constraint $g_i(x) \geq c_i$, rewrite it as $-g_i(x) \leq -c_i$. If you are asked to satisfy an equality constraint $g_i(x) = c_i$, write it as two constraints $g_i(x) \leq c_i$, and $g_i(x) \geq c_i$, and then you'll have two multipliers for these two constraints. Once you get used to adapting this basic recipe to these sorts of variations, you'll see how to shortcut these steps. For example, you can handle equality constraints with a single multiplier if you don't constrain that multiplier to be nonnegative and if you replace

the corresponding complementary slackness condition with the condition that the constraint must hold as an equality. But until you are used to such variations the basic recipe can be adapted, as long as you're careful about it.

A1.2. The recipe at work: An example

That is a pretty amazing recipe. It will work, and in just a bit I'll try to explain what it is doing and why it will work, but first let's do an example and see it in action.

A consumer consumes two commodities, wheat and candy. If w is the amount of wheat this individual consumes and c is the amount of candy (in some units that are well specified and that we won't bother with), then the consumer's utility is given by the utility function $u(w, c) = 3 \log(w) + 2 \log(c + 2)$. The consumer seeks to maximize the utility he gets from his consumption of wheat and candy, subject to four constraints. The amounts of wheat and candy consumed must both be nonnegative. The consumer has \$10 to spend, and the price of wheat and the price of candy are each \$1 per unit. A unit of wheat contains 150 calories and a unit of candy 200 calories, and the consumer is constrained to eat no more than 1550 calories.

Mathematically, the consumer's problem is to pick w and c to solve the following problem:

$$\max 3 \log(w) + 2 \log(2 + c)$$

$$\text{subject to } c \geq 0$$

$$w \geq 0$$

$$w + c \leq 10, \text{ and}$$

$$150w + 200c \leq 1550.$$

(This mathematical formulation should be obvious to you. But in case it is not: The third constraint is the budget constraint, and the fourth is the calorie constraint.)

Now to apply the recipe from the cookbook, we rewrite the first two constraints in the form needed for our recipe: $-w \leq 0$ and $-c \leq 0$. Then we

1. Form the Lagrangian.

A1.2. The recipe at work: An example

I'll use μ_w for the multiplier on the constraint $-w \leq 0$, μ_c for $-c \leq 0$, λ for $w + c \leq 10$, and ν for $150w + 200c \leq 1550$, so the Lagrangian is

$$3 \log(w) + 2 \log(2 + c) - \lambda(w + c) - \nu(150w + 200c) + \mu_w w + \mu_c c.$$

(Note that the last term is $-\mu_c(-c)$, which comes to $\mu_c c$.)

2. Write the first-order conditions for w and c .

They are

$$\frac{3}{w} - \lambda - 150\nu + \mu_w = 0, \text{ and}$$

$$\frac{2}{2 + c} - \lambda - 200\nu + \mu_c = 0.$$

3. Write the four constraints.

We've already done this, so I won't repeat it here.

4. Constrain all the multipliers to be nonnegative.

$$\lambda \geq 0, \nu \geq 0, \mu_w \geq 0, \text{ and } \mu_c \geq 0.$$

5. Write the four complementary slackness conditions.

These are

$$\mu_w w = 0, \mu_c c = 0, \lambda(10 - w - c) = 0, \text{ and } \nu(1550 - 150w - 200c) = 0.$$

6. Look for a solution (in w , c , and the four multipliers) to all the equations and inequalities above.

The way this is done, typically, is by trial and error, working with the complementary slackness conditions (hereafter, CSCs). What the CSCs tell you is that either one thing or another equals zero. For example, $\mu_w w = 0$ means that either μ_w or w is zero. Could it be w ? If $w = 0$, then the first of the first-order conditions won't be solved, since that has a $3/w$ in it. So μ_w will have to be zero, and we can disregard it in all that follows.

Next, what about $\mu_c c = 0$? Is it possible that $c = 0$? If this happens, then the second first-order condition reads $1 - \lambda - 200\nu + \mu_c = 0$. So either

$\lambda > 0$ or $\nu > 0$ or both. (Otherwise, the left-hand side of the equation just given will be at least 1; remember that multipliers are always nonnegative in this recipe.) Now by complementary slackness, the only way that λ can be positive is if $w + c = 10$, and since we're hypothesizing that $c = 0$, this will mean $w = 10$. That's not bad, since then the diet constraint is met easily: $150 \cdot 10 + 200 \cdot 0 < 1550$. But then ν will have to be zero. (Why? complementary slackness again.) And then the FOCs (first-order conditions), with $w = 10$ and $c = 0$ inserted, will read

$$.3 - \lambda = 0 \text{ and } 1 - \lambda + \mu_c = 0.$$

The first tells us that $\lambda = .3$. But then the second won't work, because μ_c will have to be $-.7$, and negative multipliers aren't allowed. So (summing up what we know so far), if $c = 0$, then λ can't be positive. But then ν will have to be positive (if $c = 0$), and that is even worse; complementary slackness says that the only way $\nu > 0$ is if $150w + 200c = 1550$, and this together with $c = 0$ gives $w = 31/3 > 10$, and the budget constraint is violated. So, we conclude $c = 0$ won't be a solution and $\mu_c = 0$ will have to hold.

Now look again at the first FOC: $3/w - \lambda - 150\nu = 0$. I've taken the liberty of getting rid of the μ_w , since we know there won't be a solution with $\mu_w \neq 0$. Since $3/w > 0$ for any positive w , one or both of λ and ν must be positive. (A similar conclusion would have emerged if we had looked at the second FOC.) So we have three cases left to try: $\lambda > 0$ (alone), $\nu > 0$ (alone), or both. Take $\lambda > 0$ and $\nu = 0$ first. Then (complementary slackness, one more time) $w + c = 10$. The FOCs become $3/w - \lambda = 0$ and $2/(2 + c) - \lambda = 0$. That is three equations in three unknowns, and we can solve $\lambda = 3/w = 3/(10 - c)$, which substituted into the second FOC is $2/(2 + c) = 3/(10 - c)$, or $(2 + c)/2 = (10 - c)/3$, or $3(2 + c) = 2(10 - c)$, or $6 + 3c = 20 - 2c$, or $5c = 14$, or $c = 14/5$, which then gives $w = 10 - 14/5 = 36/5$, and (if anyone cares) $\lambda = 15/36 = 5/12$.

Bingo! We have an answer! Or do we? The FOCs are solved. The multipliers are nonnegative. The CSCs hold. But we have to go back and check the original constraints. And, when we look at the calorie constraint ... disaster: $(36/5)(150) + (14/5)(200) = 1640$. Oh well.

Let me try $\lambda = 0$ and $\nu > 0$ next. Then by the CSC, $150w + 200c = 1550$. In addition, the FOCs now read $3/w - 150\nu = 0$ and $2/(2 + c) - 200\nu = 0$. That is three equations in three unknowns, and we can solve. First write the two FOCs as $3/w = 150\nu$ and $2/(2 + c) = 200\nu$. Then we can eliminate ν , since $3/(150w) = \nu = 2/(200(2 + c))$. Rewrite this equation as $150w/3 = 200(2 + c)/2$, or $50w = 200 + 100c$. Multiply both sides by 3,

to get $150w = 600 + 300c$, and then use this and the first equation to eliminate w , or $600 + 300c + 200c = 1550$, or $c = 950/500 = 1.9$. This gives $w = 4 + 3.8 = 7.8$. Which is nice, because this means this solution meets the budget constraint. It hits the calorie constraint on the nose (since it was designed to do so). It solves the FOCs (if we figure out what ν must be) and the CSCs. But we still need to check that the value of ν that comes out of the FOCs is nonnegative. And it is: $\nu = 1/(50w) = 1/390 = .00256410 \dots$. Let me summarize this solution:

$$w = 7.8, c = 1.9, \nu = .00256410 \dots,$$

with $\lambda = \mu_w = \mu_c = 0$.

There is actually one more thing to try, namely $\lambda > 0$ and $\nu > 0$. I'll leave it to you to show that this doesn't work with a few hints. First, if both multipliers are to be positive, then complementary slackness says that both constraints must be satisfied with equality, or $w + c = 10$ and $150w + 200c = 1550$. That is two equations in two unknowns, and you can solve for w and c directly. Then with these two values for w and c , the two FOCs become two equations in two unknowns, those unknowns being λ and ν . And you can solve those. And when you do, one of the two multipliers will turn out to be negative — no solution there. (For those readers who have seen this all before and are reading along only to review this, you know that the solution lies where the calorie constraint and not the budget constraint binds. So you should be able to tell before doing the algebra which of the two multipliers will turn out to be negative at the point where both constraints bind.)

We've solved the problem, and in a moment I'll try to give you some intuition about just why this recipe works. But before doing so, I want to make two comments about the problem. First, if we had used a little common sense, we could have saved a bit of work. This is a very typical consumer budgeting problem, with the not-so-typical addition of a calorie constraint. Nonetheless, we can draw the shape of the feasible set of wheat-candy pairs as in figure A1.

The line b-b is the budget constraint line, and c-c is the calorie constraint line, so the shaded quadrilateral is the set of feasible consumption pairs. Now this consumer is locally insatiable; his utility increases if we increase either his wheat consumption or his candy consumption or both, so the only possible places an answer could be found would be at one of the points marked x , y , and z , or on the line segment xy , or the segment yz . Each of these corresponds to a particular set of binding constraints, and